
The Accuracy of Matched Sample Surveys: Evidence from the Cooperative Election Study

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Abstract Nonprobability samples have grown in popularity among survey researchers, with proponents claiming that they can deliver accurate estimates at lower cost than probability samples. This growth has occurred despite accumulating evidence that nonprobability samples are generally less accurate. A notable exception appears in a small number of evaluations of YouGov, particularly a Pew study, which found YouGov to yield more accurate measurements than other nonprobability sample providers. Because YouGov uniquely relies on sample matching, perhaps its accuracy is attributable to that approach. This paper presents the largest evaluation of YouGov data to date, assessing accuracy across nine waves of the Cooperative Election Study, from 2006 to 2022. Across 3,927 demographic estimates, 908 turnout and voting-method estimates, and 4,705 candidate-choice estimates, the CES manifested substantial error, often quite large, even when using YouGov’s proprietary weights. Accuracy did not improve over the 17-year period. Without those weights, accuracy was considerably lower, meaning that sample matching alone yielded less accurate measurements. Conventional demographic post-stratification did not improve accuracy for variables outside the weights and slightly increased error, consistent with prior studies.

Following eye-opening embarrassments for pollsters a century ago stemming from nonprobability sampling, scholars have relied on random sampling for decades to produce highly accurate measurements. Yet with the emergence of the internet in 2000, nonprobability sampling reemerged as a dominant methodology. Some firms implemented probability sampling for online questionnaire administration, improving efficiency in data collection and processing. But other firms have collected data more inexpensively by abandoning probability sampling. These providers have routinely claimed to be implementing sophisticated methods to yield remarkably accurate measurements.

However, many studies have shown that nonprobability samples yielded less accurate measurements than probability samples (see Cornesse et al. [2020](#) for a review of 25 studies)

and in some cases, strikingly less accurate (e.g., MacInnis et al. 2018; Yeager et al. 2011). Nonetheless, some scholars and survey firms have argued that advanced nonprobability methods can yield accurate estimates (Ansolabehere and Rivers 2013; Crampton 2007), raising the possibility that many prior evaluations did not reflect the performance of more sophisticated vendors (see Baker et al. 2013).

The present study is a large-scale evaluation of the accuracy of YouGov’s nonprobability sample surveys, which have used the method of sample matching (Rivers 2007). Sample matching was used initially for causal inference; matched control groups were constructed to estimate treatment effects (Stuart 2010). Applied to internet surveys, this approach selects a sample from among nonrandom online panel members to approximate characteristics of a probability sample in terms of a small set of variables, usually demographics (Rivers 2006). YouGov (n.d.) has claimed to be one of the most quoted pollsters and market research firms in the world. The company operates in more than 55 countries and is frequently cited by global media.

We focus on nine Cooperative Election Study (CES) surveys conducted over a 17-year period by YouGov, all of which make use of sample matching. The CES, initially called the Cooperative Congressional Election Study, is the product of a collaboration of universities that contributed funding, supplemented by support from the National Science Foundation. YouGov collected data from large samples before and after each national election in the U.S. since 2006. CES data have been widely used across political science, cited over 4,700 times in scholarly publications as of August 2026.¹

We begin below by reviewing theoretical views of probability and non-probability sampling, then briefly describe the history of these methods’ use in American survey research during the last century. Building on the methodologies employed in past comparative studies of survey data accuracy, we then report assessments of the accuracy of CES measurements, broken down to isolate the respective contributions of sample matching and standard post-stratification

¹Based on a Google Scholar full-text search for “Cooperative Congressional Election Study” OR “Cooperative Election Study” that returned approximately 4,770 works.

weighting.

As we will see, this distinction is critical.

Nonprobability Sampling and the Rise of YouGov

Probability sampling selects units at random with known inclusion probabilities, ideally so that every member of a population has a known non-zero probability of becoming part of the intended sample. In theory, if members of such a sample are invited to complete a questionnaire, the result should closely resemble what would be obtained if all members of the population had done so.

When large-scale survey research began in the U.S., however, probability sampling was not the norm. Well-known researchers, including Gallup, Crossley, and Roper, disseminated results of surveys of non-probability samples, to great acclaim (Crossley 1937). Only after this methodology yielded embarrassing inaccuracy, especially in 1948 (see Mosteller et al. 1949), did probability sampling become the standard for high-quality social scientific research (Cornesse et al. 2020). This methodology has been the foundation of major U.S. survey programs for decades, including the Census Bureau’s Current Population Survey (CPS) and American Community Survey (ACS), the Bureau of Labor Statistics’ Consumer Expenditure Surveys, the CDC’s National Health Interview Survey, and others.

With the arrival of the internet, enterprising survey researchers drew probability samples and invited people to complete online questionnaires, increasing efficiency (compared to face-to-face and telephone interviewing) and affording the privacy of self-completion. But firms soon recognized that opt-in nonprobability panels offered even greater speed and cost savings. Although some observers dismissed non-probability samples as “quick and dirty” tools (Bethlehem 2010), their low per-respondent costs made such web surveys attractive to researchers (K. B. Wright 2005).

As nonprobability samples gained popularity, a substantial benchmarking literature

emerged comparing them with high-quality probability samples across domains including voting behavior, consumption, sexual attitudes, and demographics. These studies generally found that probability samples yielded more accurate estimates, whereas nonprobability samples were less accurate and more variable in their accuracy (e.g., Chang and Krosnick 2009; Dutwin and Buskirk 2017; Erens et al. 2014; MacInnis et al. 2018; Sturgis et al. 2018; Szolnoki and Hoffmann 2013; Yeager et al. 2011). Synthesizing evidence from 25 studies, Cornesse et al. (2020) concluded that “even in the age of declining response rates, accuracy in probability sample surveys is generally higher than in nonprobability sample surveys” (p. 22).

Researchers began to ask whether post-survey adjustment—especially weighting—can compensate for selection bias in opt-in panels and yield accurate population estimates. A 2013 AAPOR review concluded that such adjustments sometimes reduced bias but produced inconsistent results across outcomes and did not reliably eliminate coverage, nonresponse, and selection biases (Baker et al. 2013). Methods have evolved, though, and some studies have suggested that sophisticated techniques can reduce bias. For example, Wang et al. (2015) used multilevel regression and post-stratification to obtain accurate 2012 presidential-election estimates from a highly unrepresentative Xbox sample. In a hybrid design combining probability and nonprobability samples, Dever (2018) found that restricting propensity-score adjustment of the nonprobability sample to the region of common support reduced bias for some estimates, though not all. But other recent studies reach the opposite conclusion. Dutwin and Buskirk (2017) found that propensity weighting reduced bias in an opt-in panel but was less effective than simple raking, and adjusted nonprobability samples remained more biased than low-response-rate probability samples. Yeager et al. (2011) showed that demographic weighting improved the accuracy of some non-probability estimates but did not do so for most. Likewise, MacInnis et al. (2018) reported that demographic post-stratification consistently benefited probability samples but rarely improved nonprobability samples. Taken together, this evidence is not reassuring that conventional post-stratification and other post-survey adjustments can solve the inaccuracy of non-probability samples.

Sample size is another proposed remedy for the error in nonprobability samples. In probability samples, increasing n reduces sampling variance at a well-defined $1/\sqrt{N}$ rate. In nonprobability samples, a larger n does not necessarily reduce selection bias, coverage bias, bogus-response bias, or adjustment-model error (Elliott and Valliant 2017; Mercer, Kreuter, et al. 2017). Meng (2018) formalized this concern as the “big data paradox.” Using the 2016 CCES, the predecessor of the CES studied here, he estimated a data defect correlation of about -0.005 for self-reported support for Trump, and showed that at that level a sample of one percent of eligible voters, about 2.3 million respondents, would have the same mean squared error as a simple random sample of roughly 400. Error that is sampling noise should shrink as respondent counts grow; error that reflects who is in the sample should persist regardless of how many respondents are added. Since the CES estimates quantities for states and districts whose respondent counts vary within and across waves, we can assess, and extrapolate, whether sample matching exhibits scaling dynamics closer to that of a probability sample or other nonprobability sampling methods.

A handful of studies have examined error in sample matched surveys, specifically, and have found that matching alone provides little or no consistent advantage over conventional post-stratification or raking (Brick et al. 2015; Buskirk and Dutwin 2015; Mercer, Lau, and Kennedy 2018). Evidence from specialized populations similarly indicates that matching can improve sample comparability without necessarily eliminating selection bias (DiSogra et al. 2015).

But not all nonprobability survey designs are equivalent. YouGov emerged on the scene using a different method. They drew early attention after outperforming traditional telephone pollsters in several U.K. parliamentary elections, including the 2001 and 2005 contests (Coleman 2001; Electoral Calculus 2005). Following its 2007 acquisition of Polimetrix, YouGov entered the U.S. market and adopted Polimetrix’s sample matching methodology. Ahead of the 2008 presidential election, the firm emphasized its ability to deliver rapid, low-cost survey estimates while maintaining accuracy, positioning sample matching as a

potential solution to the limitations of opt-in panels (Bradshaw 2008; Crampton 2007).

This accuracy claim received notable support in a 2016 Pew Research Center evaluation. In a head-to-head comparison of eight nonprobability vendors and a probability sample fielded on the same questionnaire, YouGov outperformed all other opt-in samples and also exceeded the accuracy of Pew’s probability sample (Kennedy et al. 2016). YouGov exhibited the lowest measurement error across 20 benchmark variables and performed especially well among Black and Hispanic respondents, populations for which survey error is typically higher (Hopkins et al. 2024). Earlier studies likewise found YouGov samples to produce accurate estimates (Ansolabehere and Rivers 2013; Ansolabehere and Schaffner 2014; Rivers and Bailey 2009; Vavreck and Rivers 2008). Therefore, perhaps sample matching represents a substantive advance in nonprobability survey design.

However, these past evaluations did not assess the independent impacts of matching and weighting. Each study examined data where both matching and weighting had been done. The confounding is most problematic for the offices used to demonstrate accuracy. Since 2014, the CES has incorporated statewide U.S. Senate and gubernatorial election outcomes in its weights, mechanically minimizing error for precisely the races used to demonstrate accuracy. Another limitation of past studies is reliance on a single survey year, which can mislead given the higher variance of nonprobability samples (MacInnis et al. 2018). These limitations motivate a broader, systematic evaluation of YouGov’s accuracy and whether conventional demographic post-stratification improves accuracy in nonprobability matched samples independent of YouGov’s weighting choices. The investigation described in this paper presents the most comprehensive such assessment to date, using multiple election-year surveys administered by YouGov for the Cooperative Election Study.

The Cooperative Election Study

The CES describes its survey samples as representative of U.S. citizens ages 18 and older. The questionnaires have measured an array of variables, including political opinions, demographics, and candidate choices in federal and state elections, including for President, U.S. Senate and House, Governor, Attorney General, Secretary of State, State Senator, and State Representative. Table 1 reports, for each office and year, the number of states with CES respondents and the average number of respondents per state. YouGov has conducted CES data collection in collaboration with a consortium of university and organizational research teams.

Fielded in every federal election year since 2006, the CES involved two online waves of data collection each year: a pre-election survey administered before Election Day and a post-election survey administered afterward.² The exact field dates varied across years (see Table 2). All respondents who completed the pre-election wave were re-invited to the post-election wave.

Table 2 reports what the CES guides label AAPOR Response Rate 1, which divides completed interviews by all sampled cases not known to be ineligible. We label it instead as a completion rate, following AAPOR’s standard definitions for survey outcomes (American Association for Public Opinion Research 2023, p. 78), which note that the population of interest for an opt-in panel is not well defined. The number of people exposed to a recruitment invitation is unknown, and the narrower quantity that can be measured—how many invited panel members completed a given survey—should not be called a response rate, because unlike in a probability sample a high value does not imply reduced risk of nonresponse bias. Completion rates ranged from 13.9 percent in 2016 to 61.2 percent in 2020, with no monotonic trend, partly because the reporting base changed between years. The 2014 and 2016 figures pooled YouGov panelists with respondents from external vendors, whose completion rates

²This paper focuses on surveys done in federal election years. The CES has also collected data in other, odd-numbered years using a different design, and we do not consider those data.

were far lower (41.9 versus 8.2 percent in 2016). The 2018–2022 figures appear to cover YouGov panelists only. The 2018 guide says so explicitly; the 2020 and 2022 guides do not, but their disposition tables count fewer completes (60,466 and 64,957) than the pooled YouGov and vendor interviews that entered matching (74,099 and 80,233). Even among these YouGov-only years, completion rates ranged from 30.0 to 61.2 percent, so the change in reporting base does not account for all of the variation. The 2008–2012 guides do not state which respondents the rate covers. Their sampling sections name external panels alongside YouGov’s, and none of their disposition tables separates respondents by sample source. The 2008 table counts slightly fewer completes (49,043) than the roughly 50,800 pre-election interviews that entered matching, the same pattern as in 2020 and 2022; the 2010 guide reports no matching-pool size; and the 2012 table is divided by study component rather than by sample source. The participation rate from the general population was unobserved and presumably far lower.

CES Sample Matching and Weighting Procedure

Sample matching is possible when a large but unrepresentative “reservoir” of respondents, typically members of an opt-in survey panel, is available (Cochran [1977](#)). A *target sample* is drawn randomly from a large probability sample, and for each person in the target sample, a “matching” individual in the reservoir is selected, based on shared characteristics. When the reservoir is sufficiently large and diverse, and the number of variables used to identify matches is large, the matched sample can have joint distributions of the matching variables similar to those in the target sample. Rivers ([2006](#)) suggested that a matched sample can thus effectively be treated as a random sample.

The CES sample was produced in five stages. Table [4](#) defines the terms used below, and Table [3](#) reports the variables used at each stage in each year. YouGov first constructed a synthetic sampling frame from the American Community Survey (ACS) and drew a stratified target sample from it. Panelists from YouGov and other opt-in providers were then invited

Table 1: Number of States with CES Respondents and Mean Respondents per State, by Office and Year

Year	President	U.S. Senate	U.S. House	Governor	Atty. Gen.	Sec. of State	State Sen.	State Rep.
Number of States with CES Respondents								
2006	*	32	50	36	28	24	42	45
2008	51	33	12	11	-	-	39	41
2010	*	36	13	37	30	26	41	44
2012	51	33	43	11	10	7	40	43
2014	*	34	43	35	31	26	41	45
2016	51	34	43	12	10	8	41	43
2018	*	33	43	36	29	25	40	46
2020	51	34	43	11	9	7	41	44
2022	*	34	44	36	28	23	41	46
Mean Number of CES Respondents per State								
2006	*	601	487	580	502	379	405	404
2008	469	351	509	319	-	-	408	453
2010	*	784	913	807	839	676	702	732
2012	769	856	840	492	679	618	663	706
2014	*	507	738	773	734	621	626	651
2016	878	932	947	547	778	584	740	811
2018	*	940	919	900	934	747	809	826
2020	888	699	969	557	974	701	814	891
2022	*	861	868	875	1058	821	784	800

* Office not on ballot that year.

- CES did not measure vote choice for that office that year.

Note: For U.S. House, respondent counts are summed across congressional districts within each state before averaging.

Table 2: CES Sample Descriptions, 2006–2022

Year	Sample Size	Completion Rate	Pre-Election Wave	Post-Election Wave
2006	36,421	not published	October	Two weeks following Nov. 7
2008	32,800	46.8%	October	Two weeks following Nov. 4
2010	55,400	40.4%	October	Two weeks following Nov. 2
2012	54,535	33.4%	October	Two weeks following Nov. 6
2014	56,200	23.4%	October	Two weeks following Nov. 4 ^a
2016	64,600	13.9%	Sept. 28 – Nov. 7	Nov. 9 – Dec. 14
2018	60,000	30.0%	Sept. 27 – Nov. 5	Nov. 7 – Dec. 3
2020	61,000	61.2%	Sept. 29 – Nov. 2	Nov. 8 – Dec. 14
2022	60,000	47.6%	Sept. 29 – Nov. 8	Nov. 10 – Dec. 15

^a The 2014 guide gives Election Day as November 6, 2014; the correct date is November 4, 2014. The window reported here uses the correct date.

Source: Publicly-available CES data guides (<https://tischcollege.tufts.edu/research-faculty/research-centers/cooperative-election-study/data-downloads>).

Note: Guides through 2014 reported field periods only at monthly or relative precision; exact opening and closing dates were published from 2016 onward. Sample sizes refer to the released pre-election questionnaire. Completion rates are the figures reported by YouGov as AAPOR Response Rate 1; the 2006 guide reported no AAPOR outcome rates. Completion rates for 2014 and 2016 pool YouGov and external-vendor respondents; 2018 covers YouGov panelists only, and 2020 and 2022 appear to, since their disposition tables report fewer completes than the pooled pre-election interviews entering matching. The 2008–2012 guides do not identify which respondents the rate covers; the 2012 figure is for the Common Content sample. All samples were non-probability samples recruited from online panels.

to complete the questionnaire. From among those who did, a matching function selected the respondents closest to the target cases, termed the *matched sample*. Finally, the matched sample was weighted in two steps: first to the frame, by propensity score weighting or entropy balancing, and then by post-stratification to demographic and political margins.

The Sampling Frame and Target Sample

In 2006, the target sample was drawn from consumer databases; from 2008 onward, CES documentation describes the sampling frame as a stratified sample drawn from the full sample of the prior year’s ACS, with records selected within strata by weighted sampling with replacement. The variables defining these strata are not named. YouGov then imputed political and religious variables onto each frame record, using models developed from the CPS Voting Supplement, the National Election Pool, the Pew Religious Landscape Survey, and prior CES data collections; the imputation models have not been publicly disclosed.

Table 3: Variables Used in Each Stage of CES Sample Matching and Weighting

		2006	2008	2010	2012	2014	2016	2018	2020	2022
Demographic										
Age	Target strata	✓	✓	✓	✓	✓	✓	×	×	×
	Matching	×	✓	✓	✓	✓	✓	✓	✓	✓
	Frame weighting	×	✓	✓	✓	✓	✓	✓	✓	✓
	Post-stratification	×	✓	✓	✓	✓	✓	✓	✓	✓
Gender	Target strata	✓	✓	✓	✓	✓	✓	×	×	×
	Matching	×	✓	✓	✓	✓	✓	✓	✓	✓
	Frame weighting	×	✓	✓	✓	✓	✓	✓	✓	✓
	Post-stratification	×	✓	✓	✓	✓	✓	✓	✓	✓
Race	Target strata	✓	✓	✓	✓	✓	✓	×	×	×
	Matching	×	✓	✓	✓	✓	✓	✓	✓	✓
	Frame weighting	×	·	·	·	✓	✓	✓	✓	✓
	Post-stratification	×	✓	✓	✓	✓	✓	✓	✓	✓
Education	Target strata	·	✓	✓	✓	✓	✓	×	×	×
	Matching	×	✓	✓	✓	✓	✓	✓	✓	✓
	Frame weighting	×	✓	✓	✓	✓	✓	✓	✓	✓
	Post-stratification	×	✓	✓	✓	✓	✓	✓	✓	✓
Hispanic origin	Frame weighting	×	·	·	·	·	·	✓	✓	✓
Marital status	Matching	×	✓	✓	✓	·	·	·	·	·
Family income	Matching	×	✓	✓	✓	·	·	·	·	·
Employment status	Matching	×	·	·	·	✓	✓	·	·	·
Ideology	Matching	×	✓	✓	✓	✓	✓	·	·	·
	Frame weighting	×	·	·	·	✓	✓	·	·	·
Party ID (baseline)	Matching	×	✓	✓	✓	✓	✓	·	·	·
	Frame weighting	×	·	·	·	✓	✓	·	·	·
Political interest	Matching	×	✓	✓	✓	✓	✓	·	·	·
	Frame weighting	×	·	·	·	✓	✓	·	·	·
Religious tradition	Matching	×	✓	✓	✓	·	·	·	·	·
Church attendance	Matching	×	✓	✓	✓	·	·	·	·	·
“Born again” status	Matching	×	·	·	·	✓	✓	·	·	·
	Frame weighting	×	·	·	·	✓	✓	·	·	·
	Post-stratification	×	·	·	·	·	·	✓	✓	✓
Census region	Matching	×	✓	✓	✓	·	·	·	·	·
Metropolitan status	Matching	×	✓	✓	✓	·	·	·	·	·
State of residence	Post-stratification	×	·	·	·	✓	✓	✓	✓	✓
State size	Target strata	✓	·	·	·	·	·	×	×	×
District competitiveness	Target strata	✓	·	·	·	·	·	×	×	×
Took post-election wave	Matching	×	·	·	·	✓	✓	·	·	·
Interactions among the above	Matching	×	·	·	·	✓	✓	·	·	·
	Frame weighting	×	·	·	·	✓	✓	✓	✓	✓
Voting Administration										
Voter registration	Target strata	✓	·	·	·	✓	✓	×	×	×
	Matching	×	✓	✓	✓	✓	✓	✓	✓	✓
	Frame weighting	×	·	·	·	✓	✓	·	·	·
	Post-stratification	×	·	·	·	✓	✓	✓	✓	✓
Turnout, earlier election	Frame weighting	×	✓	✓	✓	·	·	·	·	✓
Candidate Choice										
Presidential vote, prior election	Frame weighting	×	·	·	·	·	·	·	·	✓
	Post-stratification	×	·	·	·	·	·	✓	✓	✓
Presidential vote, current election	Post-stratification	×	·	·	·	·	·	·	✓	·
Vote choice, governor	Post-stratification	×	·	·	·	✓	✓	✓	✓	✓
Vote choice, U.S. senator	Post-stratification	×	·	·	·	✓	✓	✓	✓	✓

Note: A check indicates that a variable was used at the specified stage; a dot indicates documented non-use. A × indicates that the guide does not document which variables were used at that stage.

Table 4: CES Sample Matching Terms

Term	Definition
Population of Interest	All U.S. citizens ages 18 and older. The population about which the CES seeks to draw conclusions.
Target Sample	A stratified random sample drawn from the ACS. Serves as the probability sample that the matched sample attempts to replicate.
Panel Reservoir	The full set of panelists from YouGov and other potential respondents who have agreed to take surveys with other companies.
Invited Respondents	Potential respondents selected for invitation to complete the pre-election questionnaire, typically chosen using a cross-classification on demographic variables.
Participating Respondents	People who completed the pre-election questionnaire (or both the pre-election and post-election questionnaires).
Matched Sample	The subset of participating respondents whose characteristics most closely matched the target sample according to YouGov’s matching function.
Frame-Weighted Sample	The matched sample after weighting to the sampling frame, by propensity score weighting (2008–2012) or entropy balancing (2014–2022).
Final Weighted Sample	The frame-weighted sample after post-stratification to demographic and political margins, trimming, and normalization. These are the weights released with the CES.

The target sample was then drawn from the frame. From 2008 through 2016, the guides describe stratification by age, race, gender, and education, with simple random sampling within strata; voter registration status was added as a stratifying variable in 2014 and 2016. The 2006 target sample was stratified on registration status, state size, and congressional district competitiveness, as well as age, race, and gender. The 2018–2022 guides continue to describe the ACS-based frame but not the draw of the target sample from it, so the target strata for those years are unknown (Table 3).

Sourcing Panel Members

YouGov maintains a large opt-in, nonprobability panel recruited in a variety of ways, including via standard advertising and strategic partnerships with websites. At registration, new panelists provided socio-demographic information. Panelists have then been invited to complete questionnaires in exchange for points redeemable for small monetary rewards at selected vendors (YouGov 2026).

For the CES, respondents were drawn both from YouGov’s panel and from potential respondents supplied by other companies (Table 5). In 2016, for example, 106,382 people completed the questionnaire, roughly half of them YouGov panelists. The CES guides state that invitations were balanced across a cross-classification of demographic variables (Appendix 4), but they do not describe how the pool of panelists from which invitees were drawn was determined.³ Rivers (2006) describes the general procedure YouGov developed for this purpose, in which invitations are allocated using a cross-classification on age, gender, race, education, region, and sample source, with the number of invitations per target case set from estimated completion probabilities so as to yield at least one completed interview per case. Whether the CES followed this procedure in a given year is not documented.

³An inquiry about the selection process has been made but has not yet been answered.

Table 5: Non-YouGov Panelist Providers

Year	External Providers
2006	None
2008	E-Rewards, Western Wats
2010	E-Rewards, Western Wats
2012	E-Rewards, Western Wats
2014	MyPoints, Research Now, Survey Sampling International
2016	MyPoints, Research Now, Survey Sampling International, Global Market Insights
2018	Dynata, Prodege, Critical Mix
2020	Dynata, Prodege, Critical Mix
2022	Dynata, Prodege, Disqo, Generation Lab

Matching

The pool of completed pre-election questionnaires exceeded the desired sample size. From it, the matched sample was constructed by selecting the N respondents who most closely resembled the N cases in the target sample (Rivers 2007). Similarity was measured with a weighted distance metric rather than by exact or propensity score matching (Stuart 2010). The distance between target case t and respondent i was

$$D(t, i) = \sum_{k=1}^K w_k d_k(t_k, i_k),$$

where d_k measures dissimilarity on matching variable k and w_k sets that variable’s importance, with larger weights demanding closer agreement. For unordered categorical variables, YouGov specified distance matrices between pairs of categories.

The guides for 2008 through 2016 print the matching function in full. In these years, matching drew on political as well as demographic information; the variables always included gender, age, race, education, party identification, ideology, and political interest; church attendance and region were also used in 2008–2012, and employment status in 2014–2016 (Table 3). From 2018, the guides describe the function in prose rather than printing it, and the matching variables are demographic only. The 2006 matched sample was selected from more than 150,000 PollingPoint panelists, but the guide does not name the matching

variables.

Matching brings the joint distribution of the matching variables in the matched sample close to that of the target sample, but it does not constrain any margin explicitly. Remaining imbalance is addressed by the two weighting steps.

Weighting, Step 1: Weighting to the Frame

The first weighting step adjusted the matched sample toward the sampling frame. From 2008 to 2012, this was done by propensity score weighting. A logistic regression predicted membership in the frame, as against the matched sample, from age, education, gender, and turnout in the previous election; for frame records, turnout was the CPS-reported value attached to the synthetic record. Matched respondents were grouped into deciles of the estimated propensity score and weighted to the frame’s distribution across those deciles. The 2006 guide does not describe a weighting procedure.

From 2014 through 2022, the matched sample was weighted to the frame by entropy balancing (Hainmueller [2012](#)). This method assigns each matched respondent a weight such that the weighted sample reproduces specified moments of the frame while departing as little as possible from uniform weights. In 2014 and 2016, the balancing conditions included political as well as demographic variables. In 2018 and 2020 they were purely demographic, and in 2022 presidential vote was added.

Weighting, Step 2: Post-Stratification

The frame-weighted sample was then post-stratified. In every documented year, the adjustment included age, gender, education, and race. Voter registration status was added in 2014, and born-again status and prior presidential vote were added in 2018. Weights for the common content were additionally adjusted across states, to the outcomes of that year’s gubernatorial and Senate elections from 2014 onward, and to the presidential vote in 2020. Where these outcomes also serve as benchmarks, their agreement with weighted CES

estimates partly reflects the weighting itself, a point we return to in the results. Weights were trimmed at 7 in the propensity-weighting years and at 15 from 2014 onward, and final weights were normalized to the sample size.

Benchmarks

Demographics

Population targets were drawn primarily from Current Population Survey (CPS) data provided by IPUMS (Flood et al. [2024](#)). The CPS is a monthly probability sample survey conducted by the U.S. Census Bureau since 1940. Sampled households have been interviewed once a month for four months, then given four months off, and again once a month for four additional months. Initial interviews were conducted in person, and subsequent interviews were conducted in person or by telephone. Interviewers attempted to collect data from all household members and accepted proxy reports from one household member about another household member when necessary. Annual CPS household response rates in the survey years studied ranged from 72.9 percent (2022) to 92.0 percent (2010) (Table [B2](#)) (U.S. Bureau of Labor Statistics [n.d.](#)). CPS data are weighted to account for sampling design, nonresponse, and undercoverage, using population controls derived from the decennial census, which are updated annually.

Election Returns and Turnout

State and federal election returns were compiled for all 50 states and the District of Columbia from 2006 to 2022. Presidential election returns were obtained from the MIT Election Data and Science Lab (MIT Election Data and Science Lab [2017](#)); U.S. House and Senate returns were obtained from the Federal Election Commission; gubernatorial returns were obtained from CQ Press; and returns for other statewide offices were obtained from

Ballotpedia. Voter turnout data were drawn from the University of Florida Election Lab (McDonald [n.d.](#)). Any remaining data were obtained through correspondence with state election offices.

Methods

Measuring Error

CES error was assessed by comparing survey estimates to external benchmarks. For demographics, turnout, and voting method, error was computed as the difference between the CES-estimated share of the modal category and the benchmark share of that category, where the modal category is the one with the largest share in the benchmark. For example, because women make up the largest share of the adult population, we compared the CES-estimated female share to the benchmark female share.

With the exception of U.S. House elections, the unit of analysis is the state. For each statewide race, we compared the CES estimate of vote share in the state to the certified share from that state’s official election returns. For U.S. House contests, each respondent was assigned to a congressional district based on address, and error was computed at the district level.

Vote choice was measured analogously: we compared the CES-estimated share of the vote won by the modal candidate—the candidate who received the most votes in the actual election—to that candidate’s certified vote share in the official returns. Prior CES evaluations have measured vote-choice accuracy using Democratic vote share. We instead use the vote share of a specific candidate wherever possible.

The wording of the CES vote-choice question changed over time. In earlier years respondents were asked which *party* they voted for, and in later years which *candidate*, with party labels provided alongside candidate names. In reviewing election returns, we identified numerous cases in which party-level reporting could not be cleanly mapped to survey

responses, including contests with multiple candidates sharing a party label, candidates appearing on multiple party lines, ballots without party labels, and candidates switching party affiliation between the primary and general election. Thus, candidate-level comparison avoids mismeasurement that party-level aggregation can introduce, and we prefer candidate-level returns whenever possible and relied on party-level comparisons only when candidate-level information was unavailable. Candidate vote choice was available for all presidential contests in all years, for U.S. Senate and House elections beginning in 2008, for all gubernatorial elections beginning in 2010, and for attorney general and secretary of state races beginning in 2018. State legislative contests were evaluated using party vote share in all years.

Below, we report the root mean squared error (RMSE):

$$\text{RMSE}_{v,w} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{P}_{\text{CES},i,w} - P_{\text{true},i})^2},$$

where v indexes the benchmark variable being evaluated (e.g., the proportion of college graduates, Donald Trump’s vote share, etc.); $\hat{P}_{\text{CES},i,w}$ is the CES estimate for a variable v in state (or, for congressional races, district) i under weighting scheme w ; $P_{\text{true},i}$ is the corresponding benchmark value; and N is the number of distinct benchmark comparisons included in the calculation (i.e., the number of errors). Table 6 lists the number of error observations for each variable and year.

RMSE penalizes large deviations more heavily than small ones, making it a useful summary measure of survey accuracy: surveys that are consistently close to benchmark values across states have lower RMSE than those with similar average accuracy but occasional large errors. Following MacInnis et al. (2018) and Yeager et al. (2011), we report error separately for variables used to construct survey weights, since these are explicitly constrained to match population benchmarks. Accordingly, we distinguish between *primary* variables, those used in either step of the CES weighting (weighting to the frame or post-stratification), and *secondary* variables, which were not. Because the CES weighting scheme changed across waves, status is assigned separately for each variable in each year (see Table 3). When the

Table 6: Number of Error Observations by Variable and Year

Variable	2006	2008	2010	2012	2014	2016	2018	2020	2022	Sum
<i>Demographic</i>										
Age Group	51	51	51	51	51	51	51	51	51	459
Sex	51	51	51	51	51	51	51	51	51	459
Hispanic Origin	-	-	51	51	51	51	51	51	51	357
Education	51	51	51	51	51	51	51	51	51	459
Family Income	51	51	51	51	51	51	51	51	51	459
Employment Status	51	51	51	51	51	51	51	51	51	459
Union Membership	51	51	51	51	51	51	51	51	51	459
Veteran Status	51	51	51	51	51	51	51	51	51	459
Residence Duration	-	51	51	51	51	51	-	51	51	357
<i>Voting Administration</i>										
Turnout	50	50	50	50	50	50	50	50	50	450
Voting Method	50	51	51	51	51	51	51	51	51	458
<i>Candidate Choice</i>										
President	*	51	*	51	*	51	*	51	*	204
U.S. Senate	32	33	36	33	34	34	33	34	34	303
U.S. House	50	132	152	426	422	426	424	427	429	2,888
Governor	36	11	37	11	35	12	36	11	36	225
Attorney General	28	-	30	10	31	10	29	9	28	175
Secretary of State	24	-	26	7	26	8	25	7	23	146
State Senator	42	39	41	40	41	41	40	41	41	366
State Representative	45	41	44	43	45	43	46	44	46	397
Total Errors per Year	714	816	926	1,181	1,194	1,185	1,142	1,184	1,197	9,539

* Office not on ballot that year.

- CES did not measure that variable that year.

Note: Shows number of state-level observations (errors) for each variable by year. Washington D.C. is counted as the 51st state.

weights include a prior-election measure, such as previous turnout, we treat the corresponding current-election variable as primary in that year (turnout in 2008–2012 and 2022). President in 2020 is primary because the post-stratification weights targeted the 2020 presidential vote itself. Beginning in 2014, statewide U.S. Senate and gubernatorial outcomes were used in constructing the weights, as was the 2020 presidential vote, so estimates for these offices in those years are not independent tests of accuracy.

Weighting Scenarios

Comparing CES accuracy across weighting scenarios allows us to separate the contributions of sample matching and post-survey adjustment. Because sample matching is intended to approximate a random sample, error in the unweighted CES reflects the performance of matching alone, while differences between weighted and unweighted estimates capture the incremental accuracy provided by weighting.

CES weights reflect design choices specific to YouGov, including the incorporation of known U.S. Senate and gubernatorial outcomes. To assess whether conventional demographic post-stratification improves accuracy in nonprobability matched samples independent of these choices, we evaluated accuracy under an alternative weighting scheme applied consistently across years.

To this end, we generated an additional set of weights designed to minimize divergence between the CES and a set of demographic and turnout benchmarks drawn from the CPS and administrative sources. Only CES variables with strictly comparable CPS counterparts—identical or fully collapsible response categories measuring the same underlying construct—were eligible for inclusion in the weighting process. Notably, race was excluded because CPS race categories could not be exhaustively mapped to CES categories. As a result, although 29 CES variables had conceptual CPS counterparts across years, only 9–11 per year were used for weight generation (Table 7).

Weights were then constructed following the procedures proposed by DeBell and Krosnick

Table 7: Variables Used during ANESRake Weight Generation

Variable	2006	2008	2010	2012	2014	2016	2018	2020	2022
Age Group	✓	✓	✓	✓	✓	✓	✓	✓	✓
Education	✓	✓	✓	✓	✓	✓	✓	✓	✓
Employment Status	✓	✓	✓	✓	✓	✓	✓	✓	✓
Family Income	✓	✓	✓	✓	✓	✓	✓	✓	✓
Hispanic Origin	–	–	✓	✓	✓	✓	✓	✓	✓
Sex	✓	✓	✓	✓	✓	✓	✓	✓	✓
Union Membership	✓	○	✓	✓	✓	✓	✓	✓	✓
Veteran Status	✓	✓	✓	✓	✓	✓	✓	✓	✓
Turnout	✓	✓	✓	✓	✓	✓	✓	✓	✓
Voting Method	✓	✓	✓	✓	✓	✓	✓	✓	✓
Residence Duration	–	✓	✓	✓	✓	○	–	✓	✓

- ✓ Variable used as a target for ANESRake weighting.
- Variable provided to ANESRake but not selected by the algorithm.
- Variable not provided to ANESRake (benchmark unavailable for that year).

(2009) using anesrake (Pasek 2018). In order from the most to least discrepant category, categories whose weighted CES proportions differed from CPS benchmarks by at least 5 percent were updated sequentially. At each step, current weights were multiplied by the ratio of the population proportion to the weighted sample proportion. Weights were capped at 5 and re-centered to maintain a mean of 1. Iteration continued until no benchmark category exceeded the 5 percent tolerance or until convergence.

Results

We first document the error in CES estimates produced by sample matching alone. We then ask how much the CES weights reduce that error, and whether the reduction depends on how a variable entered the design: through matching, through weighting, or through neither. Next, we ask whether accuracy changed across the seventeen-year period when data were collected, and we document the distribution of errors to show how often CES errors were sizable. We then ask whether increasing sample size is associated with decreased error, as statistical theory anticipates for probability samples. Finally, we ask whether error is smaller in blow-out races than in knife-edge races, and whether conventionally computed

post-stratification weights perform differently than the weights YouGov provides with CES data.

Error with Matching Only

Table 8: Matching-Only RMSE by Variable and Year

Variable	2006	2008	2010	2012	2014	2016	2018	2020	2022	Mean	SD
<i>Demographic</i>											
Age Group	7.8	6.4	12.4	9.8	7.6	8.8	7.6	8.9	8.4	8.6	1.7
Sex	5.2	5.4	3.3	3.6	4.0	6.1	5.3	6.2	5.6	5.0	1.1
Hispanic Origin	-	-	7.4	8.9	8.4	8.3	9.2	9.3	5.3	8.1	1.4
Education	5.6	3.9	13.8	7.7	5.0	7.3	6.4	6.1	7.1	7.0	2.8
Family Income	4.3	3.8	5.2	3.4	3.3	4.0	4.0	4.9	7.0	4.5	1.2
Employment Status	11.0	7.3	10.6	15.7	12.5	10.8	12.5	12.5	11.8	11.6	2.2
Union Membership	5.5	3.7	5.4	5.3	5.2	4.8	4.9	5.5	4.8	5.0	0.6
Veteran Status	8.7	7.2	11.2	9.4	7.0	5.6	5.4	4.5	4.6	7.1	2.3
Residence Duration	-	4.2	10.4	8.8	5.8	6.1	-	5.0	5.2	6.5	2.2
Demographic Mean	6.9	5.3	8.8	8.1	6.5	6.9	6.9	7.0	6.7	7.0	1.0
<i>Voting Administration</i>											
Turnout	47.7	30.5	43.8	37.8	43.5	35.4	40.4	30.8	40.1	38.9	5.9
Voting Method	4.3	4.0	4.7	6.2	3.9	5.3	5.0	8.5	8.0	5.5	1.7
Voting Administration Mean	26.0	17.3	24.3	22.0	23.7	20.4	22.7	19.7	24.0	22.2	2.7
<i>Candidate Choice</i>											
President	-	5.8	-	4.4	-	8.1	-	10.1	-	7.1	2.5
U.S. Senate	5.6	8.0	6.1	4.3	5.9	7.3	5.9	9.9	9.9	7.0	1.9
U.S. House	4.3	10.7	9.3	7.7	8.6	8.1	8.1	10.1	11.4	8.7	2.1
Governor	4.9	4.8	6.0	2.8	4.8	7.1	5.8	11.0	10.9	6.4	2.8
Attorney General	12.3	-	12.6	15.4	9.3	6.5	7.1	11.3	12.8	10.9	3.1
Secretary of State	11.5	-	6.7	9.7	9.0	10.0	7.7	11.0	12.5	9.8	1.9
State Senator	10.1	5.9	9.7	8.0	13.4	9.4	7.5	13.4	14.6	10.2	3.0
State Representative	7.0	5.4	6.2	5.7	8.7	8.6	7.2	11.7	13.6	8.3	2.8
Candidate Choice Mean	8.0	6.8	8.1	7.3	8.5	8.1	7.0	11.1	12.2	8.6	1.9
Yearly Mean	9.7	7.3	10.3	9.2	9.2	8.8	8.8	10.0	10.8	9.4	1.0
Excluding Turnout	7.2	5.8	8.3	7.6	7.2	7.3	6.8	8.9	9.0	7.6	1.0

- Variable not measured in the CES that year.

Note: Means average the variable-level RMSEs in each year. Yearly Mean is the average of all variable RMSEs for that year. SDs are the spread of the yearly values across waves.

Table 8 reports CES accuracy across all available measures using YouGov’s matched sample, but *not* the CES weights. Pooling RMSE estimates across the nine survey waves yields 9,540 individual error observations. The average RMSE across waves was 9.4 percentage points, ranging from 7.2 in 2008 to 11.1 in 2022; excluding turnout, the average was 7.6.

Accuracy varied by variable class. Demographic variables had the lowest average error (6.9), followed by candidate choice (8.7), while voting-administration measures had far higher error (22.1), driven by turnout, whose RMSE averaged 38.7 and exceeded 30 points in every wave.

Among demographic measures, the CES exhibited the highest error for employment status (11.6) and the lowest for family income (4.4). Among offices, error was lowest for governor (6.5), U.S. Senate (7.0), and president (7.1), and highest for attorney general (11.0), secretary of state, and state senator (both 10.5).

Demographic error was stable across waves, with the class average varying by about one point. Turnout error varied the most ($SD = 5.9$). Candidate-choice error was steady through 2018, between 6.7 and 8.8 points, and then rose to 11.1 in 2020 and 13.1 in 2022; the CES weights largely removed the 2020 increase (see below).

Error with Matching and CES Weights

Table 9 reports error under the weights YouGov provides, which reflects the error users of the CES would typically encounter.

The weights reduced average error from 9.4 to 7.9 percentage points, and from 7.6 to 6.1 when turnout is excluded. Substantial error remains in the version of the survey most researchers use. Turnout was still in error by 37.1 points on average, and employment status remained the least accurate demographic measure (11.8). The largest improvements were in candidate choice, whose average fell from 8.7 to 6.7; in 2020, it fell from 11.1 to 4.6.

These averages combine variables that were used to construct the weights with variables that were not. The remaining tables separate the two. They classify each variable separately in each year by whether it was used in matching, in either weighting step, or in neither, and pool the underlying estimates across the years in which it held that status. Because the 2006 documentation does not permit this classification, these tables cover 2008–2022.

Table 9: Matching + Weighting RMSE by Variable and Year

Variable	2006	2008	2010	2012	2014	2016	2018	2020	2022	Mean	SD
<i>Demographic</i>											
Age Group	7.4	4.5†	5.3†	3.8†	4.3†	3.0†	3.3†	3.1†	3.0†	4.2	1.4
Sex	5.4	6.9†	5.0†	1.6†	2.2†	1.2†	3.0†	2.8†	3.0†	3.5	1.9
Hispanic Origin	-	-	7.7	8.7	8.0	8.3	8.7†	8.2†	3.8†	7.6	1.7
Education	13.2	3.2†	6.4†	3.6†	4.0†	5.8†	4.1†	4.4†	3.9†	5.4	3.1
Family Income	4.6	3.5	3.9	3.8	4.5	4.4	5.2	6.6	8.6	5.0	1.6
Employment Status	7.2	10.8	10.7	14.0	12.0	12.8	13.3	13.9	12.1	11.9	2.1
Union Membership	5.3	3.7	5.6	5.9	5.1	5.1	5.7	6.3	5.3	5.3	0.7
Veteran Status	4.9	4.7	5.3	4.5	4.6	4.2	3.2	3.3	3.4	4.2	0.8
Residence Duration	-	6.0	7.5	4.9	5.3	5.1	-	4.2	6.2	5.6	1.1
Demographic Mean	6.9	5.4	6.4	5.6	5.6	5.5	5.8	5.8	5.5	5.8	0.5
<i>Voting Administration</i>											
Turnout	44.3	26.5†	31.8†	38.8†	47.3	36.8	35.4	26.7	48.3†	37.3	8.2
Voting Method	3.7	4.5	6.0	4.8	3.5	5.5	5.0	6.8	6.3	5.1	1.1
Voting Administration Mean	24.0	15.5	18.9	21.8	25.4	21.2	20.2	16.7	27.3	21.2	3.9
<i>Candidate Choice</i>											
President	*	3.5	*	2.9	*	1.7	*	2.3†	*	2.6	0.8
U.S. Senate	6.0	5.1	2.0	2.4	3.1†	4.0†	4.5†	3.3†	4.8†	3.9	1.3
U.S. House	4.9	9.4	11.4	9.8	9.1	9.6	8.4	8.0	9.9	9.0	1.8
Governor	4.9	4.7	3.5	3.4	3.5†	4.0†	4.2†	4.3†	5.7†	4.2	0.8
Attorney General	13.2	-	11.8	15.2	8.0	4.9	4.2	3.6	7.5	8.6	4.4
Secretary of State	11.1	-	6.4	10.9	8.2	9.6	4.2	3.7	6.5	7.6	2.9
State Senator	10.7	5.5	9.7	7.5	12.1	8.0	4.5	6.4	8.8	8.1	2.5
State Representative	7.3	4.9	5.4	6.4	6.3	6.5	4.6	5.5	8.5	6.1	1.2
Candidate Choice Mean	8.3	5.5	7.2	7.3	7.2	6.0	4.9	4.6	7.4	6.5	1.3
Yearly Mean	9.6	6.7	8.1	8.0	8.4	7.4	7.1	6.5	8.6	7.8	1.0
Excluding Turnout	7.3	5.4	6.7	6.3	6.1	5.8	5.4	5.4	6.3	6.1	0.7

† primary variable that year (used in CES weighting).

The 2006 CES guide does not record which variables entered matching or weighting, so primary and secondary status cannot be confirmed.

* Office not on ballot that year.

- Variable not measured in the CES that year.

Note: Means average the variable-level RMSEs in each year. Yearly Mean is the average of all variable RMSEs for that year. SDs are the spread of the yearly values across waves.

Matching Status, CES Weights, and Error

Table 10: RMSE Before and After Weighting for Variables Used and Not Used in Matching

Variable	Matching-Only	Matching + Weighting	Δ RMSE
<i>Variables Used in Matching</i>			
Age Group	8.9	3.9	-5.0
Sex	5.0	3.7	-1.4
Education	7.7	4.5	-3.2
Family Income	4.2	3.7	-0.5
Employment Status	11.7	12.4	+0.7
Mean	7.5	5.6	-1.9
<i>Variables Not Used in Matching</i>			
Hispanic Origin	8.2	7.8	-0.4
Family Income	4.8	6.1	+1.2
Employment Status	12.0	12.5	+0.5
Union Membership	5.0	5.4	+0.4
Veteran Status	7.2	4.2	-3.0
Residence Duration	6.8	5.7	-1.1
Turnout	38.1	37.3	-0.8
Voting Method	5.9	5.4	-0.5
President	7.4	2.7	-4.7
U.S. Senate	7.4	3.8	-3.6
U.S. House	9.2	9.3	+0.1
Governor	7.2	4.3	-2.9
Attorney General	10.9	8.6	-2.2
Secretary of State	9.4	7.0	-2.4
State Senator	10.7	8.1	-2.6
State Representative	8.9	6.1	-2.8
Mean	9.9	8.4	-1.6

Note: RMSE is calculated across all applicable error observations from the 2008–2022 waves and reported in percentage points. Status is assigned by variable and year, so variables that change status may appear in both sections. Δ RMSE equals Matching + Weighting minus Matching-Only; negative values indicate lower error after weighting. The 2006 wave is excluded because matching and weighting classifications are unavailable.

Table 10 groups variables by whether they were used in matching. Before weighting, variables used in matching had lower error than those that were not (7.5 versus 10.1), although much of this gap reflects turnout, which was never a matching variable. Matching on a variable did not ensure that it was measured accurately: employment status was a matching variable yet had an RMSE of 11.7, and the weights made it slightly worse.

The CES weights reduced error by similar amounts in both groups (1.9 points for matched

variables, 1.5 for the rest), suggesting that variable inclusion in the matching process was less consequential than variable inclusion in the weighting process.

Weighting Status, CES Weights, and Error

Table 11: RMSE Before and After Weighting for Variables Used and Not Used in Weighting

Variable	Matching-Only	Matching + Weighting	Δ RMSE
<i>Variables Used in Weighting</i>			
Age Group	8.9	3.9	-5.0
Sex	5.0	3.7	-1.4
Hispanic Origin	8.2	7.2	-0.9
Education	7.7	4.5	-3.2
Turnout	38.4	37.2	-1.1
President	10.1	2.3	-7.7
U.S. Senate	8.0	4.0	-4.0
Governor	7.9	4.5	-3.4
Mean	11.8	8.4	-3.3
<i>Variables Not Used in Weighting</i>			
Hispanic Origin	8.2	8.2	-0.1
Family Income	4.6	5.3	+0.7
Employment Status	11.9	12.5	+0.6
Union Membership	5.0	5.4	+0.4
Veteran Status	7.2	4.2	-3.0
Residence Duration	6.8	5.7	-1.1
Turnout	37.8	37.3	-0.5
Voting Method	5.9	5.4	-0.5
President	6.3	2.8	-3.5
U.S. Senate	6.3	3.4	-2.9
U.S. House	9.2	9.3	+0.1
Governor	5.3	3.7	-1.6
Attorney General	10.9	8.6	-2.2
Secretary of State	9.4	7.0	-2.4
State Senator	10.7	8.1	-2.6
State Representative	8.9	6.1	-2.8
Mean	9.7	8.3	-1.3

Note: RMSE is calculated across all applicable error observations from the 2008–2022 waves and reported in percentage points. Status is assigned by variable and year, so variables that change status may appear in both sections. Δ RMSE equals Matching + Weighting minus Matching-Only; negative values indicate lower error after weighting. The 2006 wave is excluded because matching and weighting classifications are unavailable.

Weighting should, by design, reduce error for the variables used to construct the weights. The more informative test is whether those improvements extend to variables outside the

weights, as they generally would in a probability sample. Table 11 groups variables by whether they were used in either weighting step. Variables that changed status across years appear twice, with each row pooling only the years in which the variable held the indicated status. Negative values of Δ RMSE indicate that weighting reduced error.

Weighting reduced RMSE by 3.4 percentage points for variables used in the weights, compared with 1.3 points for variables that were not, a gain roughly 2.6 times as large. All eight rows for weighting variables improved. Among the 16 rows for non-weighting variables, 12 improved and four got worse: family income, employment status, union membership, and U.S. House vote.

Variables that switched status show the same pattern. Hispanic origin, turnout, president, U.S. Senate, and governor each improved more in the years they were weighting variables than in the years they were not; for president, the reduction was 7.7 points versus 3.4. Turnout improved little under either status (1.2 versus 0.6 points), which is consistent with how turnout entered the design: where it was used, it was prior-election turnout in the frame weighting, never the survey-year turnout we benchmark against. Because turnout error is so large, it also pulls up the average for weighting variables, which is why both groups average 8.4 after weighting despite the difference in improvement. For governor and Senate from 2014, and for president in 2020, some of the gain in weighting years is mechanical, because the weights were post-stratified to the same election outcomes that serve as benchmarks.

Error for Variables Used in Neither Matching nor Weighting

The cleanest test of whether the CES design improves accuracy beyond its own targets comes from variables that entered neither matching nor weighting in a given year (Table 12). For these, weighting reduced average error by 1.3 points, from 9.8 to 8.5.

The gains were concentrated in candidate choice. In Table 12, which pools each office's error across years, every office except the U.S. House improved, by between 1.6 and 3.4 points. Weighted error for president (2.8) and U.S. Senate (3.4) was comparable to their

Table 12: RMSE Before and After Weighting for Variables Used in Neither Matching nor Weighting

Variable	Matching-Only	Matching + Weighting	Δ RMSE
<i>Variables Used in Neither Matching nor Weighting</i>			
Hispanic Origin	8.2	8.2	-0.1
Family Income	4.8	6.1	+1.2
Employment Status	12.0	12.5	+0.5
Union Membership	5.0	5.4	+0.4
Veteran Status	7.2	4.2	-3.0
Residence Duration	6.8	5.7	-1.1
Turnout	37.8	37.3	-0.5
Voting Method	5.9	5.4	-0.5
President	6.3	2.8	-3.5
U.S. Senate	6.3	3.4	-2.9
U.S. House	9.2	9.3	+0.1
Governor	5.3	3.7	-1.6
Attorney General	10.9	8.6	-2.2
Secretary of State	9.4	7.0	-2.4
State Senator	10.7	8.1	-2.6
State Representative	8.9	6.1	-2.8
Mean	9.7	8.4	-1.3

Note: RMSE is calculated across all applicable 2008–2022 error observations for variable-years used in neither matching nor weighting and reported in percentage points. Δ RMSE equals Matching + Weighting minus Matching-Only; negative values indicate lower error after weighting. The 2006 wave is excluded because matching and weighting classifications are unavailable.

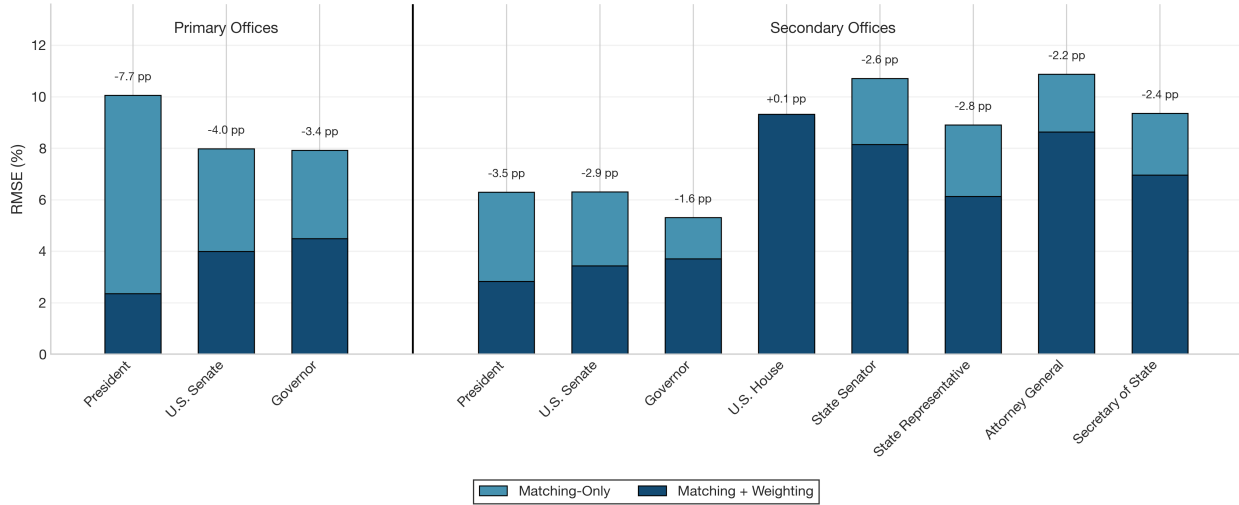


Figure 1: RMSE Reduction from CES Weighting by Office

Bars show matching-only RMSE and the RMSE remaining after CES weighting, pooled over all error observations from the 2008–2022 waves, as in Tables 11 and 12. An office appears among primary offices in the years it was a CES weighting target and among secondary offices otherwise. The 2006 wave is excluded because matching and weighting classifications are unavailable.

Alt text: Bar chart comparing error, in percentage points, before and after weighting for eight offices, shown separately for years when each office was a weighting target and years when it was not. In target years, weighting reduced error for President, United States Senate, and Governor by 7.7, 4.0, and 3.5 points. In other years, reductions were smaller, from 1.6 to 3.4 points, and United States House error rose slightly.

error in years when they were weighting targets (2.4 and 4.1), although the two sets of years differ. Demographic variables outside the design mostly did not improve: family income, employment status, and union membership got worse, and Hispanic origin was essentially unchanged. Veteran status was the exception, improving by 3.0 points. Turnout improved by 0.6 points and remained in error by 37.1.

Separating the effects of weighting by office, as in Figure 1, reveals a similar descriptive pattern: offices included in the weights generally exhibit lower overall error and larger apparent reductions in error attributable to weighting.

These results suggest that the CES weights provided some benefits beyond the variables used to construct them, but those benefits were substantially smaller and less consistent, and they were concentrated in candidate choice, not demographics.

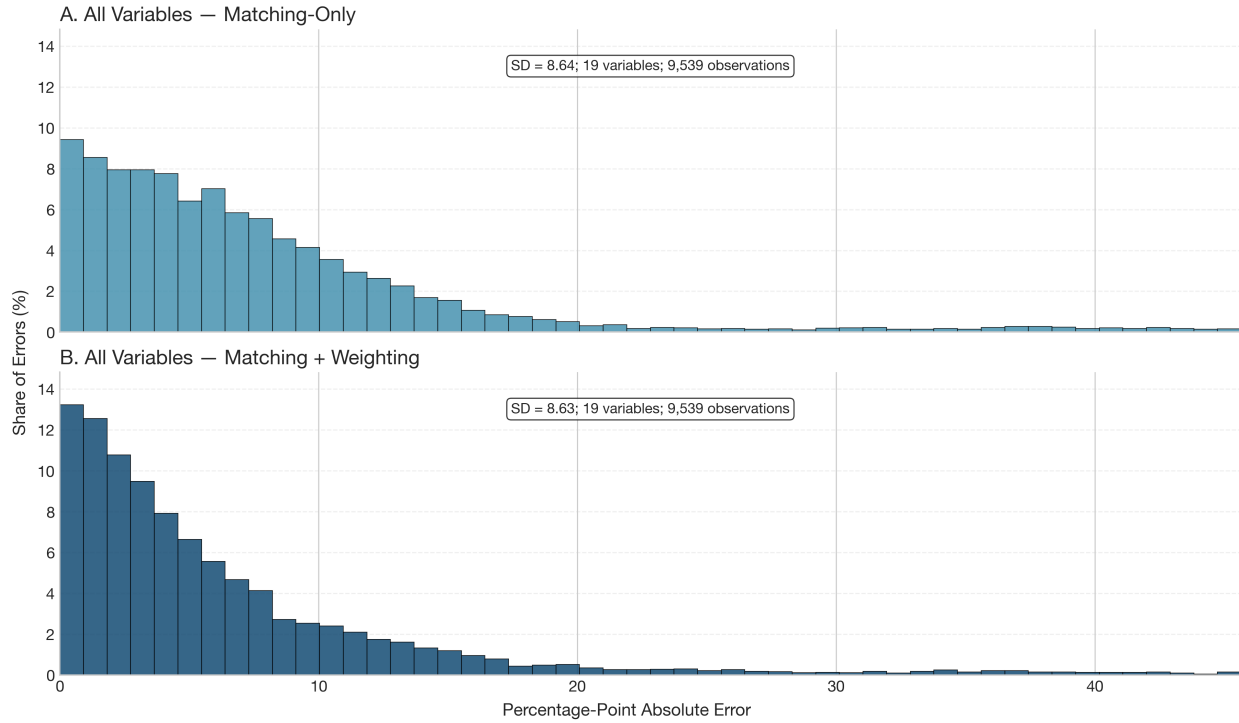


Figure 2: Distribution of CES Absolute Error Before and After Weighting

Each histogram shows the share of all errors at each level of percentage-point absolute error, pooling 9,540 observations across 19 variables and nine CES waves. Panel A reports matching-only estimates; Panel B reports estimates using YouGov’s weights. Weighting concentrated more estimates near their benchmarks but left the overall dispersion of errors essentially unchanged ($SD = 8.66$ and 8.64 , respectively).

Alt text: Two histograms, one above the other, of absolute error for all 9,540 estimates before and after weighting. Before weighting, the share of estimates declines gradually from zero error, with many estimates between 2 and 10 percentage points. After weighting, estimates cluster more tightly near zero, but both distributions keep a long, flat tail of errors above 20 points.

Distribution of Errors

Prior work has shown that a hallmark of nonprobability samples is wide variation across measures in the amount of error observed. It has further shown that post-stratification, which corrects known imbalances in probability samples, produces smaller and less consistent gains in nonprobability ones (Chang and Krosnick 2009; Lavrakas et al. 2022; MacInnis et al. 2018; Mercer, Lau, and Kennedy 2018; Walker, Pettit, and Robinson 2009; Yeager et al. 2011).

Figure 2 shows both patterns across the 2006–2022 CES waves. Under matching alone, absolute error was not concentrated near zero: 10.2% of estimates were within one percentage point of their benchmark, 28.3% were within three points, and 55.4% missed their benchmark

by more than five points. The CES weights shifted the distribution toward the origin, increasing the share of estimates within three points from 28.3% to 39.6%. Mean absolute error declined from 8.00 to 6.87 percentage points, and the median declined from 5.71 to 4.06 points. The standard deviation of absolute error, however, was nearly unchanged at 8.66 points under matching alone and 8.64 points after weighting, suggesting that weighting improved the typical estimate without producing a comparable contraction in the extreme tail.

The effect of applying the CES weights was smaller for larger errors. The proportion of errors exceeding 15 points declined only from 11.3% to 10.2% after weighting, while the proportion exceeding 20 points changed from 6.56% to 6.36%. In total, 5.23% of estimates exceeded 20 points under both methods. Of the 626 estimates whose matching-only errors exceeded 20 points, 127 moved below that threshold after weighting and 499 remained above it. At the same time, 108 estimates that had previously been below 20 points crossed above the threshold, leaving 607 weighted estimates above it. The CES weighting procedure therefore changed the composition of the extreme-error tail but produced little change in its overall frequency.

Error Across Time

Advocates of nonprobability samples have sometimes argued that their accuracy has improved over the years as opt-in online panels expanded, recruitment methods matured, and post-stratification and weighting procedures became more sophisticated. The CES provides an opportunity to test this claim: across survey waves, it has repeatedly revised its weighting scheme and drawn respondents from multiple panel providers.

Contrary to this expectation, CES accuracy did not improve over the 17 years studied. Rather, there was a slight, marginally significant increase in error. In a variable fixed-effects regression predicting year-by-variable RMSE—including only variables that were secondary in all years and observed in every survey wave, with the undocumented 2006 wave treated as

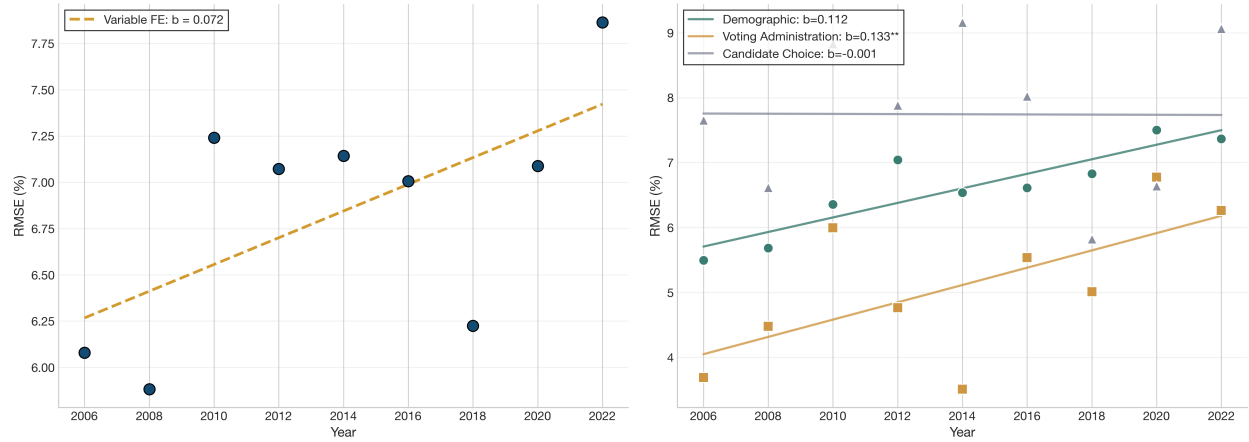


Figure 3: Secondary Variable RMSE Trends Over Time

RMSE trends over time for secondary variables that were observed in every survey wave and classified as secondary in all years. Estimates use Matching + CES Weighting. Eight variables are included: Employment Status, Family Income, Union Membership, Veteran Status, State Representative, State Senator, U.S. House, and Voting Method. **Panel A:** Each point represents the average RMSE across included variables in a given year. The dashed line shows the estimated linear trend from a variable fixed-effects regression estimated on year-by-variable RMSE, with HC3 heteroskedasticity-robust standard errors. The estimated coefficient indicates the average change in RMSE percentage points per calendar year, net of baseline differences across variables. **Panel B:** RMSE trends disaggregated by variable class. Reported coefficients come from the same fixed-effects specification estimated separately within each class, except for Voting Administration, which contains only one variable and therefore reduces to a year trend for Voting Method. Significance indicated by * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Alt text: Two scatterplots of error, in percentage points, by survey year from 2006 to 2022, with fitted trend lines. The left panel shows average error across eight variables, from about 6 points in 2006 to about 8 in 2022, with a gently rising trend. The right panel splits the variables into three groups: error for demographic variables and for voting method rises over time, while candidate choice error fluctuates around a flat line.

secondary (Figure 3, Panel A)—RMSE increased by approximately 0.075 percentage points per year, a marginally statistically significant trend (HC3 robust SE = 0.045, $p = 0.092$; 95% CI: [-0.012, 0.163]). A permutation test that randomly reassigned year labels supported the same conclusion ($p = 0.066$). The point estimate corresponds to roughly 0.15 RMSE points per survey wave, or an estimated increase of approximately 1.2 RMSE points from 2006 to 2022.

Figure 3 Panel B disaggregates these trends by variable class, and Table 13 reports model results. Demographic variables show a positive, marginally significant increase in error over time (0.112 RMSE points per year; SE = 0.061; $p = 0.067$). The single voting administration variable, Voting Method, shows a larger and statistically significant upward trend (0.135; SE

= 0.048; $p = 0.005$). Candidate-choice variables show no temporal trend in error (0.006; SE = 0.092; $p = 0.949$).

Table 13: Secondary Variable RMSE Trends Over Time

Model	β	SE	p -value	Perm. p	N_{vars}
<i>Panel A: Fixed Effects Model Across All Variables</i>					
Variable FE	0.0722	0.0445	0.105	0.083	8
<i>Panel B: Fixed Effects Models Within Variable Class</i>					
Demographic	0.1121	0.0609	0.066	–	4
Voting Administration	0.1332**	0.0476	0.005	–	1
Candidate Choice	-0.0014	0.0914	0.988	–	3

Note: Estimates use Matching + CES Weighting. β represents the change in RMSE, in percentage points, per calendar year. Panel A reports OLS with variable fixed effects, controlling for baseline RMSE differences across the consistently secondary variables; its permutation p -value is based on 10,000 permutations of year labels. Panel B reports the same fixed-effects specification estimated separately within each variable class; where a class contains only one variable, the model reduces to a year trend for that variable. All models use HC3 heteroskedasticity-robust standard errors.

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Sample Size and Estimation Error

According to statistical theory, the error attributable to sampling variability should decline in proportion to $1/\sqrt{N}$ (Cochran 1977). That is, as sample size grows, error shrinks nonlinearly, with diminishing returns: the expected improvement in accuracy from increasing sample size becomes minimal after even a few hundred units have been sampled from the population.

Absent assumptions sufficient to identify and adjust for selection, simply increasing the size of a nonprobability sample does not necessarily reduce total error, because selection and other nonsampling biases need not diminish with sample size (Baker et al. 2013; Bethlehem 2010; Meng 2018). Nevertheless, the ability to obtain large samples quickly and cheaply has repeatedly been presented as an advantage of online nonprobability samples (Baker et al. 2013, p. 12; YouGov 2012, p. 24).

We leverage variation in the number of CES respondents underlying each estimate to

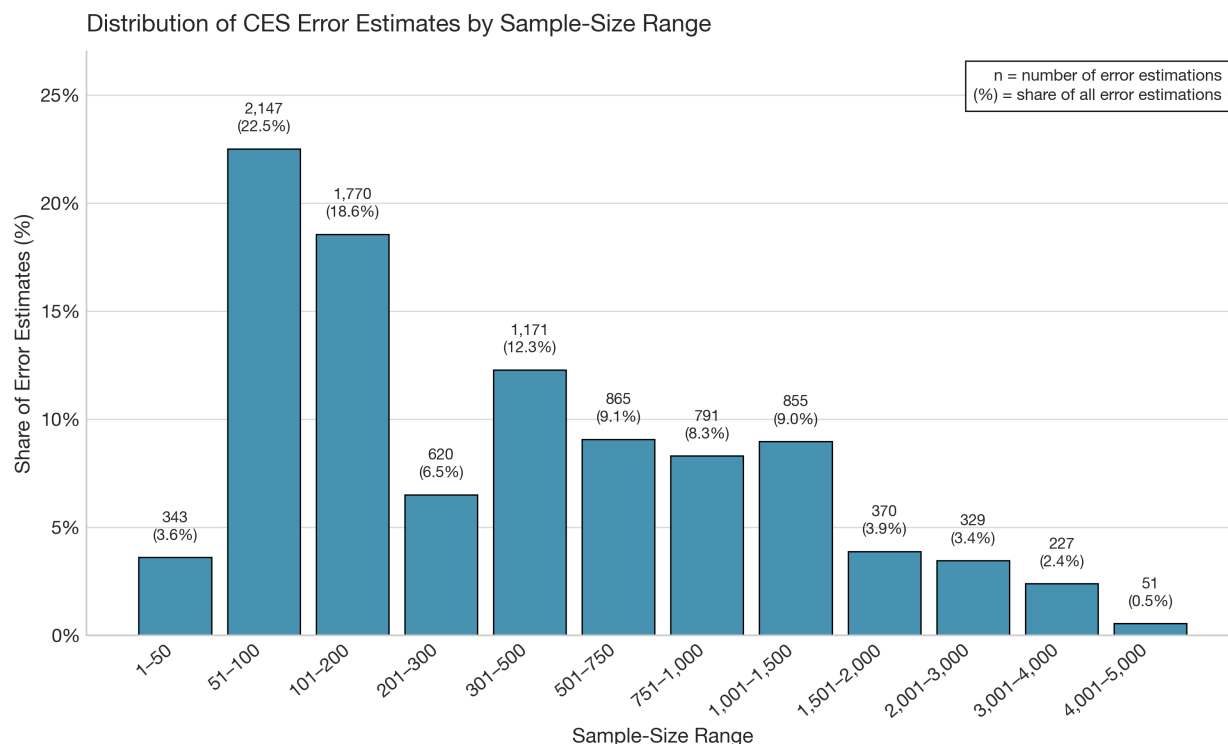


Figure 4: Distribution of CES Error Observations by Sample-Size Range

Bars report the distribution of sample sizes. Each observation is an estimate-specific comparison between a CES estimate and an external benchmark. The same underlying respondents can therefore contribute multiple error observations because the CES sample is used to estimate multiple variables. Sample size is the number of respondents contributing to the relevant estimate: the state-year respondent count for statewide estimates and the district-year respondent count for U.S. House estimates. Results use unweighted CES estimates.

Alt text: Bar chart of the share of error estimates in each of 12 sample size ranges, from 1 to 5,000 respondents. Nearly two-thirds of estimates rest on 500 or fewer respondents. The 51 to 100 range is the largest, with 22.3 percent, followed by 101 to 200, with 18.2 percent. Each range above 1,500 respondents holds less than 5 percent.

assess whether estimation error declines with sample size. Nearly two-thirds of the estimates are based on fewer than 500 respondents, but sample sizes range from 8 to 4,883, sufficient variation to examine observed error as a function of sample size (Figure 4).

Figure 5 provides a descriptive visualization of the relationship between error and sample size for variables other than turnout. It plots observed RMSE across sample-size bins and overlays a curve of the form

$$\text{RMSE}(N) = a + \frac{b}{\sqrt{N}}.$$

The curve is constrained to begin at the observed RMSE for the first bin, approximately 10.8 percentage points at $N = 50$. Under this constraint, a is estimated from the binned data and b is implied by the common starting point. Because $b/\sqrt{N} \rightarrow 0$ as N increases, the fitted curve approaches a at large sample sizes, making a a descriptive residual-error floor for these matched, nonprobability samples.

Observed RMSE initially declined with sample size, but the fitted relationship flattened at approximately 6.2 percentage points. This suggests that, whereas the probability-sampled curve continues toward zero, sample size in these nonprobability matched samples reduces error only up to a point.

Error as a Function of Contest Competitiveness

Beyond sample size, CES error also varied with contest competitiveness. As the winning candidate's benchmark vote share moved farther from 50 percent—that is, as races became less competitive—RMSE generally increased, even though sampling variance alone, which is largest at an even split, predicts the reverse. The increase was most pronounced in the least competitive races, including contests where a major-party candidate dropped out or only one major-party candidate appeared on the ballot. In these cases the benchmark winning share could approach or exceed 90 percent, while the survey-implied Democratic/Republican split remained much closer to parity.

Table 14 and Figure 6 trace this pattern by recalculating RMSE while progressively

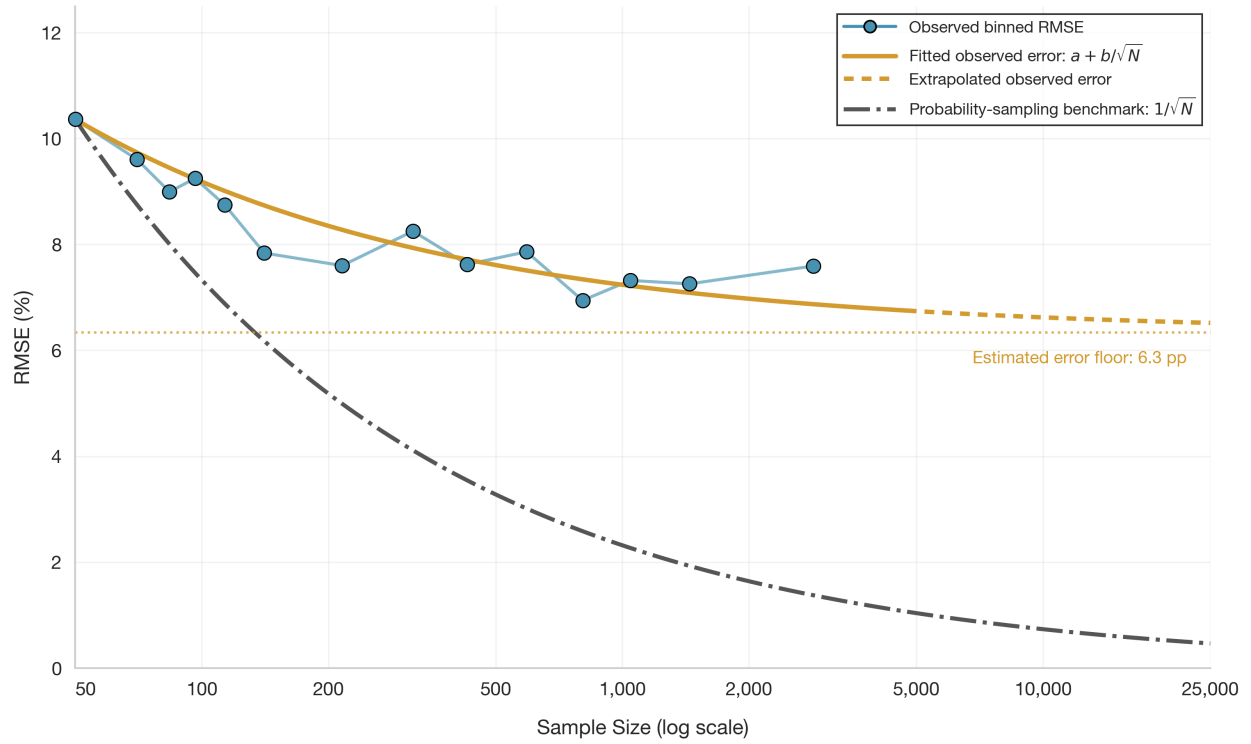


Figure 5: Observed and Extrapolated RMSE by Sample Size Relative to the Probability-Sampling Benchmark

Points show observed RMSE across sample-size bins for all variables except turnout. Bins are sample-size quantiles of roughly 650 error observations each. Each point is a single RMSE over all observations in its bin, pooled across waves and variables. The solid gold curve shows the fitted relationship $a + b/\sqrt{N}$ over the observed sample-size range, while the dashed gold segment extrapolates that relationship beyond the available data to $N = 25,000$. The gray dash-dot curve shows the theoretical $1/\sqrt{N}$ decline expected under probability sampling, calibrated to the same RMSE at $N = 50$. The dotted horizontal line marks the fitted residual-error floor. Results use unweighted CES estimates. Turnout is excluded from this visualization because it has substantially higher error.

Alt text: Line chart of error against sample size on a logarithmic scale. Observed error falls from about 11 percentage points at 50 respondents to about 8 points by 150 respondents, then stays roughly flat up to about 3,000 respondents. A fitted curve levels off at an estimated floor of 6.2 points, while the curve expected under probability sampling keeps falling toward zero.

retaining races with higher winning vote shares. For most offices, RMSE is relatively stable through moderate cutoffs but rises once high winning-share races are included, most visibly for Attorney General, Secretary of State, and State Senator contests.

Table 14: Sensitivity of Office-Level RMSE to Less Competitive Races

Office	Maximum Winning Vote Share Retained				
	$\leq 60\%$	$\leq 70\%$	$\leq 80\%$	$\leq 90\%$	$\leq 100\%$
President	2.5 (142)	2.7 (197)	2.7 (200)	2.7 (200)	2.7 (204)
U.S. Senate	4.0 (185)	4.0 (278)	4.1 (301)	4.1 (301)	4.1 (303)
U.S. House	8.7 (1,081)	9.0 (2,104)	9.1 (2,582)	9.2 (2,740)	9.3 (2,888)
Governor	4.1 (166)	4.1 (213)	4.4 (225)	4.4 (225)	4.4 (225)
Attorney General	5.2 (103)	5.1 (151)	5.9 (162)	6.0 (164)	9.5 (175)
Secretary of State	5.2 (92)	5.5 (131)	6.0 (139)	6.7 (143)	7.8 (146)
State Senator	6.0 (223)	6.5 (317)	7.5 (356)	7.7 (364)	8.5 (366)
State Representative	5.4 (261)	5.9 (365)	6.3 (395)	6.3 (397)	6.3 (397)
Mean	5.1	5.4	5.7	5.9	6.6

Note: Each cell reports RMSE in percentage points, with the number of races included in parentheses. Columns recalculate office-level RMSE after retaining only races where the winning candidate’s benchmark vote share is less than or equal to the listed cutoff. Moving from left to right progressively includes less competitive races. Estimates use Matching + CES Weighting. RMSE is pooled across the 2006–2022 waves rather than averaged by year, so values differ from the yearly means in Table 9.

This compression toward parity contrasts with prior work, which has documented systematic overstatement of winners, incumbents, and frontrunners in both post-election reports and pre-election polls (Chen et al. 2026; Hopkins 2009; Mattei 1998; G. C. Wright 1993). Among the possible mechanisms for this are that respondents may have reported their party’s candidate rather than a crossover vote for a strong winner, less salient contests may have invited weaker recall, and turnout overreporting added vote choices from people who did not vote.

U.S. House races let us examine sample size and competitiveness jointly. Because the unit of inference was the district and many districts contained relatively few CES respondents, House contests combined both sources of error in a way that state-level estimates for other offices did not. Table B3 presents a two-way RMSE matrix that stratifies U.S. House races by the winner’s vote share and by minimum district-level respondent thresholds, showing how competitiveness and district sample size interact.

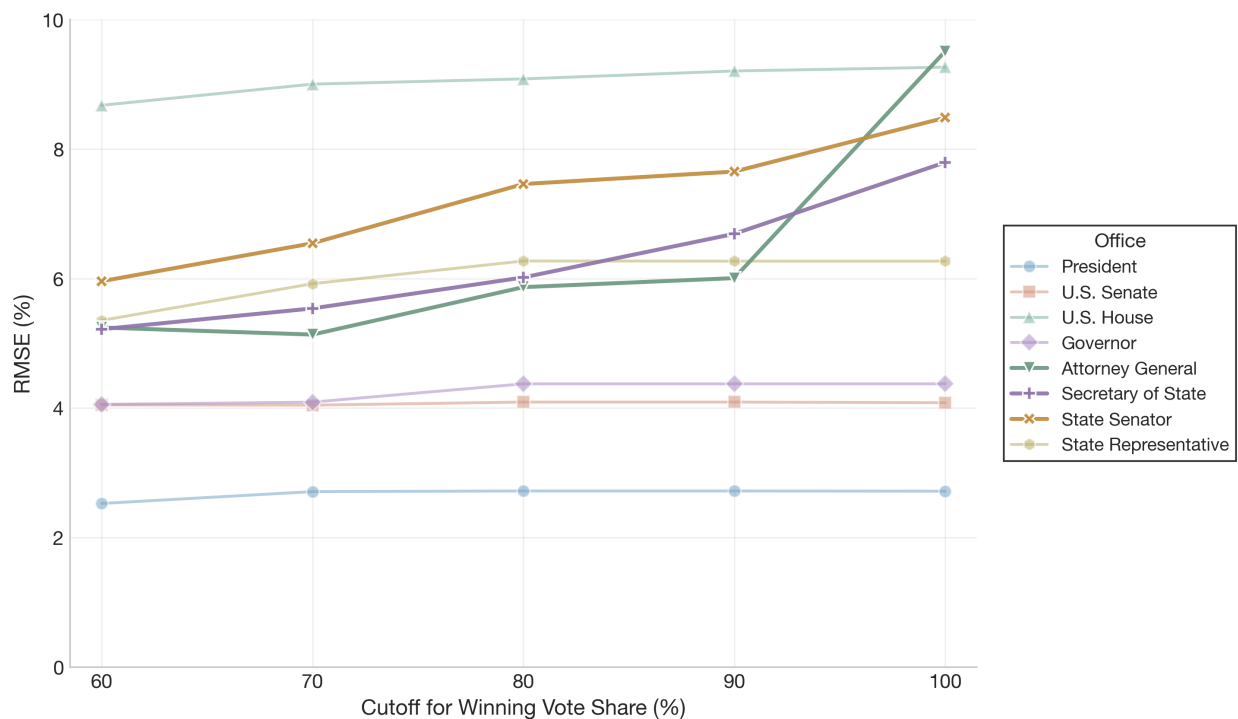


Figure 6: Sensitivity of Office-Level RMSE to Less Competitive Races

Each point recalculates office-level RMSE using only races where the winning candidate's benchmark vote share is at or below the x-axis cutoff. For example, 100 percent includes all contests for that office in the RMSE calculation, whereas 80 percent excludes races where the winning candidate received more than 80 percent of the vote. Moving from left to right progressively includes less competitive races. Error generally increases as less competitive contests are retained, with the sharpest increases occurring after very high winning-share races are included.

Alt text: Line chart of error, in percentage points, for eight offices as races with larger winning margins are progressively included, from a cutoff of 60 percent of the vote to 100 percent. Error stays nearly flat for President, United States Senate, Governor, and United States House, and rises modestly for State Representative. It rises more for State Senator, Attorney General, and Secretary of State, sharply for the last two once races won with more than 90 percent are included.

Comparison to Prior Evaluations of Error in the CES

CES documentation reported accuracy metrics for selected offices and years, though these are narrower in scope and do not distinguish between primary and secondary variables. Table 15 compares the RMSEs reported in the guides with those computed in this study.

Our RMSEs are generally higher than the reported values. The difference is small for President (+0.3 points on average) and negligible for U.S. Senate (-0.1), but larger for Attorney General (+4.0) and Secretary of State (+3.0). Averaged across years, our estimates exceed the reported values by 1.5 points.

We cannot determine with certainty what accounts for this difference, but can name a few possible sources. First, as mentioned in the methods section, the guides measured error in the Democratic share of the two-party vote, whereas we compare the modal candidate's share of all votes cast with the corresponding CES share. This number should be identical in cases where only one Democratic candidate ran, which is most cases, but nonetheless candidate-level comparisons register errors that party-level aggregation can offset, such as misestimation of independent or minor-party candidates. Second, from 2018 onward the CES guides computed error among validated voters using validated-voter weights, and earlier guides did not clearly specify the weights used. Third, the 2006 and 2010 guides include only states with at least 200 respondents, whereas we apply no minimum, so our estimates absorb more sampling error in small states. Finally, certified vote shares for the same race vary somewhat across sources, so benchmark values may differ.

Effect of Conventional Post-Stratification on CES Error

Finally, we evaluated how closely standard best-practice post-stratification applied to the nonprobability sample approached the accuracy achieved by YouGov's bespoke weighting scheme. We constructed a single national set of calibration weights using `anesrake`. To ensure a fair comparison, we evaluated accuracy only on variables that were used in neither

Table 15: RMSEs Reported in CES Data Guides and Calculated in This Study

Office		2006	2008	2010	2012	2014	2016	2018	2020	2022	Mean
President	This Study	-	3.5	-	2.9	-	1.7	-	2.3	-	2.7
	Data Guide	-	2.4	-	3.3	-	1.6	-	-	-	2.4
	Δ	-	+1.1	-	-0.4	-	+0.2	-	-	-	+0.3
Governor	This Study	4.9	4.7	3.5	3.4	3.5	4.0	4.2	4.3	5.7	4.1
	Data Guide	2.2	3.2	2.2	-	4.0	4.0	4.1	1.9	-	3.1
	Δ	+2.7	+1.5	+1.2	-	-0.5	-0.1	+0.1	+2.4	-	+1.0
U.S. Senate	This Study	6.0	5.1	2.0	2.4	3.1	4.0	4.5	3.3	4.8	3.8
	Data Guide	4.3	-	1.3	3.7	3.4	7.4	4.9	-	2.4	3.9
	Δ	+1.7	-	+0.7	-1.3	-0.3	-3.3	-0.4	-	+2.4	-0.1
Attorney General	This Study	13.2	-	11.8	15.2	8.0	4.9	4.2	3.6	7.5	10.3
	Data Guide	4.3	-	5.0	16.9	-	-	-	1.9	3.1	6.3
	Δ	+8.9	-	+6.8	-1.7	-	-	-	+1.7	+4.4	+4.0
Secretary of State	This Study	11.1	-	6.4	10.9	8.2	9.6	4.2	3.7	6.5	7.1
	Data Guide	4.4	-	4.5	9.9	-	-	3.4	5.3	4.4	5.3
	Δ	+6.7	-	+1.9	+1.0	-	-	+0.8	-1.7	+2.0	+1.8
Mean, Overlapping Offices	This Study	8.8	4.1	5.9	7.8	3.3	3.3	4.3	3.9	6.3	5.3
	Data Guide	3.8	2.8	3.3	8.4	3.7	4.3	4.1	3.1	3.3	4.1
	Δ	+5.0	+1.3	+2.7	-0.6	-0.4	-1.1	+0.1	+0.8	+2.9	+1.2

- Data not available for that year-office combination.

Note: All RMSEs are computed using CES-provided weights. “This Study” shows RMSEs calculated in this analysis; “Data Guide” shows RMSEs reported in the CES data guides. Δ = This Study – Data Guide; positive values indicate that this study calculated higher error than the data guide reported. All means, including the Mean column, use only office-years reported in both sources.

YouGov’s weighting nor anesrake post-stratification.

As a validation of the anesrake procedure, we assessed accuracy on the variables used to construct the calibration weights. Averaged across years, RMSE on these variables was at most 0.1 percentage points (Table B1); the largest single value was 0.6 (Education and Employment Status in 2010), indicating that the calibration successfully aligned the weighted sample with national benchmarks. However, this accuracy did not extend to variables excluded from both weighting schemes—State Representative, State Senator, and U.S. House races (Figure 7). For these outcomes, anesrake generally performed worse than both matching alone and matching plus the CES weights, with the partial exception of 2020 and 2022, where it improved on matching alone but still underperformed YouGov’s weights. As reported in Table 16, while YouGov’s weights reduced RMSE by an average of 1.3 percentage points across the three variables not used for weighting, anesrake weights increased error by 1.2 percentage points.

Thus, while traditional post-stratification effectively reduced error for variables directly included in the weighting scheme, it did not generally improve accuracy for other outcomes in the nonprobability matched sample, replicating prior findings that best-practice demographic weighting alone does not generalize to unconstrained variables in nonprobability surveys (MacInnis et al. 2018; Yeager et al. 2011).

Discussion

This study provides the most comprehensive evaluation to date of the accuracy of matched-sample surveys. Across nine waves of the CES and 9,540 comparisons with external benchmarks, error averaged 9.4 percentage points under matching alone and 7.9 with the CES weights. Accuracy did not improve over the 17 years studied; for variables never used in the weights, error, if anything, rose slightly, although this trend is only marginally significant. The absence of improvement is notable because the period spanned several changes of panel

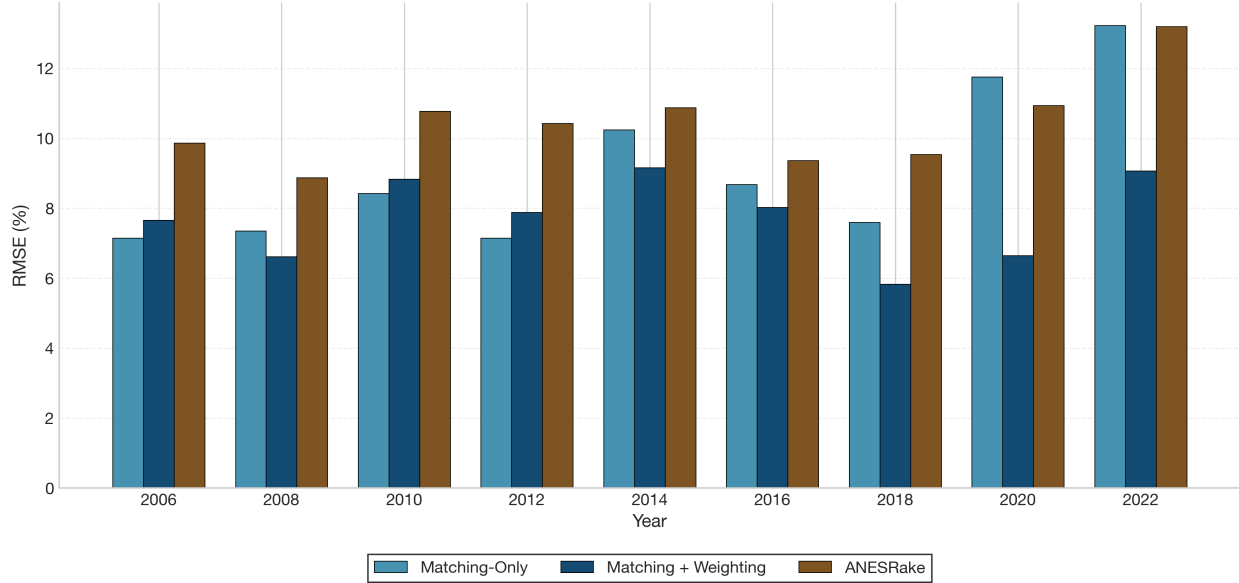


Figure 7: Comparison of Weighting Techniques Across Years

Each bar is the mean RMSE across three variables in one wave, computed from the same respondents under three weighting scenarios: matching only, with no weights; matching plus the weights YouGov releases with the CES; and matching plus national calibration weights we constructed with anesrake to CPS and administrative demographic and voting benchmarks (Table 7). The three variables, U.S. House, State Senator, and State Representative vote choice, are the only ones observed in every wave and used in neither the CES weights nor anesrake, so neither weighting scheme was fit to them.

Alt text: Grouped bar chart of error, in percentage points, by survey year from 2006 to 2022, for the three weighting approaches. The anesrake weights produced the highest error in seven of nine years. The Cooperative Election Study weights lowered error in most years, most sharply in 2020, from 11.7 to 6.6 points, and in 2022, from 13.2 to 9.1 points.

Table 16: RMSE Changes by Weighting Method

Variable	2006	2008	2010	2012	2014	2016	2018	2020	2022	Mean
<i>Matching-Only RMSE (%)</i>										
State Representative	7.0	5.4	6.2	5.7	8.7	8.6	7.2	11.7	13.6	8.3
State Senator	10.1	5.9	9.7	8.0	13.4	9.4	7.5	13.4	14.6	10.2
U.S. House	4.3	10.7	9.3	7.7	8.6	8.1	8.1	10.1	11.4	8.7
Mean	7.1	7.4	8.4	7.1	10.2	8.7	7.6	11.8	13.2	9.1
<i>Δ to Matching + Weighting (pp)</i>										
State Representative	+0.3	-0.6	-0.8	+0.7	-2.4	-2.1	-2.6	-6.2	-5.1	-2.1
State Senator	+0.6	-0.4	-0.1	-0.5	-1.3	-1.4	-3.0	-7.0	-5.8	-2.1
U.S. House	+0.7	-1.3	+2.1	+2.1	+0.5	+1.6	+0.3	-2.1	-1.5	+0.3
Mean	+0.5	-0.7	+0.4	+0.7	-1.1	-0.7	-1.8	-5.1	-4.2	-1.3
<i>Δ to Matching + ANESRake (pp)</i>										
State Representative	+3.3	+1.5	+1.3	+3.1	-1.1	+0.2	+1.7	-2.5	-0.3	+0.8
State Senator	+2.3	+1.1	+1.9	+2.8	-0.2	-0.0	+0.7	-2.8	-0.2	+0.6
U.S. House	+2.6	+2.0	+3.8	+4.0	+3.3	+1.8	+3.4	+2.8	+0.3	+2.7
Mean	+2.7	+1.5	+2.4	+3.3	+0.6	+0.7	+1.9	-0.8	-0.0	+1.4

Note: Δ equals RMSE under the indicated weighting minus Matching-Only RMSE, in percentage points; negative values indicate that weighting reduced error. The three variables shown were the only variables present in every wave from 2006 through 2022 and not used in either YouGov weighting or ANESRake.

providers and substantial revisions to YouGov’s matching and weighting procedures. The most extensive of these, in 2018, moved political information out of matching and the first weighting step and into post-stratification; it changed where that information enters the procedure, but the trend analysis shows no corresponding gain in accuracy.

A central finding is that the accuracy of the CES depends more on YouGov’s weights than on sample matching. The theoretical case for matching is that, if selection into the panel is ignorable given the matching variables, a matched sample can be treated as a random sample (Rivers 2006, 2007). In the CES, variables outside matching were estimated with substantial error, which suggests that the matching variables did not fully account for how respondents differed from the population (Mercer, Kreuter, et al. 2017). Matching also did not ensure accuracy for the variables it used: employment status was a matching variable in 2014 and 2016 yet remained among the least accurate measures in the study, and whether a variable was used in matching made little difference to how much the weights improved it. In practice, matching in the CES appears to function less as a substitute for random selection than as a

first-stage adjustment whose shortcomings the weights must then correct.

The CES weights, in turn, improved accuracy most for the variables used to construct them. Their benefits did extend beyond those targets, but selectively: among variables used in neither matching nor weighting, candidate choice improved for nearly every office, while income, employment, and union membership did not improve and in some cases got worse. This pattern is consistent with prior work showing that weighting improves an outcome mainly insofar as the outcome is related to the weighting variables (Mercer, Lau, and Kennedy 2018). The CES weights draw on political information, including presidential vote and, from 2014, statewide election results, which may explain why their benefits reached other vote choices but not demographic characteristics. By contrast, conventional post-stratification with anesrake was effective at aligning the sample with the national benchmarks it constrained, reproducing them almost exactly, but it did not improve accuracy for outcomes outside the weighting set and increased error for the three offices excluded from both weighting schemes. This result replicates a broader pattern observed in prior work: demographic post-stratification does not generalize well to unconstrained variables in nonprobability samples (Dutwin and Buskirk 2017; MacInnis et al. 2018; Yeager et al. 2011). The comparison should be read with the caveat that our anesrake weights were national, whereas the CES weights also adjust across states.

These findings align with, and in some respects extend, a growing literature documenting the limits of nonprobability survey accuracy. Prior studies have shown that average error in nonprobability samples is often only modestly worse than in probability samples, but that nonprobability designs tend to exhibit substantially greater variance, including a nontrivial share of large errors (Chang and Krosnick 2009; Cornesse et al. 2020; MacInnis et al. 2018; Yeager et al. 2011). Our results are consistent with this pattern. Under matching alone, more than half of all estimates missed their benchmarks by more than five points. The weights improved the typical estimate, but they left the dispersion of errors essentially unchanged, and roughly one estimate in ten remained in error by more than 15 points after weighting.

For a researcher analyzing a single state or office, the relevant risk is therefore not the average error but the chance that a particular estimate is badly wrong. Errors were also largest in the least competitive races, where CES estimates were compressed toward an even split, the opposite of the overstatement of winners documented in earlier election surveys (Mattei 1998; G. C. Wright 1993). If, as seems plausible in low-salience contests, reported vote choice partly reflects respondents' partisanship rather than their actual votes, CES measures of down-ballot vote choice may understate candidate-specific voting, such as split-ticket voting and support for strong candidates of the opposing party.

The results on sample size point in the same direction. In probability samples, the standard error of an estimate falls in proportion to $1/\sqrt{n}$, so error from sampling variability approaches zero as samples grow (Cochran 1977). In the CES, error declined as the number of respondents behind an estimate increased, but it leveled off at about 6.2 percentage points for variables other than turnout. The fitted curve implies that most of the reducible error is gone early: by roughly 500 respondents, predicted error is within about 1.5 points of the floor. This is the pattern expected when the remaining error is bias, which larger samples do not remove (Bethlehem 2010; Meng 2018). For comparison, a simple random sample estimating a proportion near one half would have an expected error of 6.2 points at about 65 respondents. The distinction matters because large samples are exactly what nonprobability designs make cheap (Baker et al. 2013), which raises the question of whether a large nonprobability sample is preferable to a smaller probability sample of the same population.

Nonprobability sampling is now a permanent feature of the survey research landscape. YouGov occupies a prominent position in academic and applied research, and its use of sample matching has been presented as a solution to the limitations of opt-in panels (Crampton 2007; Rivers 2007). Our evidence challenges that interpretation. Sample matching, as implemented in the CES, does not appear to deliver accuracy gains that distinguish it meaningfully from other nonprobability methods. Instead, its accuracy depends heavily on weights that incorporate outcome-adjacent information, such as prior vote and election

results, an approach that is unavailable for many research applications that lack benchmarks. For those applications, our results for variables used in neither matching nor weighting are the more relevant guide to expected accuracy.

This distinction also bears on how the accuracy of the CES has been assessed. Prior evaluations, including those in the CES guides, have largely documented accuracy through comparisons with election outcomes (Ansolabehere and Rivers 2013; Ansolabehere and Schaffner 2014; Rivers and Bailey 2009; Vavreck and Rivers 2008). We reproduce lower error estimates in some cases as well. However, we show that these conclusions are sensitive to which variables are evaluated and whether those variables were used in weighting. From 2014 onward, statewide Senate and gubernatorial results were themselves used to construct the weights, so agreement with those outcomes is not independent evidence of sample quality. Consistent with this, our estimates and those reported in the guides agree most closely for Senate and diverge most for attorney general and secretary of state, which were never weighting targets. Computational choices, including how vote shares are defined and which respondents and states are included, account for some of the remaining difference. The distinction between variables used in the weights and variables outside them is crucial for interpreting CES performance, and it has been largely absent from prior evaluations.

Several limitations qualify these conclusions. First, because the CES weighting scheme changed across waves, variables had to be classified year by year from documentation that is incomplete in places: the 2006 wave could not be classified and its variables are treated as secondary where the distinction is required, and the 2012 classifications rely on that guide’s repetition of the 2010 description. Only a limited set of variables fell outside the weights in every wave, many of them candidate-choice measures. Second, measurement of candidate choice is itself imperfect. Special elections, fusion voting, and changes in ballot structure sometimes make it difficult to determine which contest a respondent had in mind, and our candidate-level comparisons and choice of benchmark sources may produce somewhat larger errors than party-level alternatives. Third, self-reported turnout is overstated in probability

surveys as well, through overreporting and the underrepresentation of nonvoters (DeBell, Krosnick, et al. 2020; Hammer, Banks, and White 2014), so part of the turnout error is not specific to nonprobability designs; for this reason we also report results excluding turnout. These limitations reflect real constraints faced by researchers using CES data, and we do not believe they account for the broader patterns we observe.

Our results point toward several directions for research and practice. The high variance of nonprobability samples implies that single-year evaluations, still common in the literature, can be misleading; multi-year designs are essential for assessing stability, trends, and the frequency of large errors. Evaluations of nonprobability surveys should report error separately for variables inside and outside the weights, and producers of matched-sample surveys could support this by documenting the variables used at each stage of each wave. Understanding why weighting that reproduces national margins fails to improve state- and district-level estimates may shed light on the underlying sources of error. Finally, comparative evaluations across vendors remain rare; the Pew Research Center’s comparison of nonprobability vendors (Kennedy et al. 2016) is one of the few that includes YouGov, and additional work of this kind would be valuable.

In sum, our findings challenge the view of sample matching as a solution to the accuracy problems of nonprobability surveys. Combined with YouGov’s weights, matched samples perform best on the variables the weights target and somewhat better on related political measures, but they remain subject to the constraints that characterize nonprobability designs: bias that larger samples do not remove, a persistent share of large errors, and limited generalizability to outcomes the weights do not target. Researchers using CES data should interpret results with caution, particularly for variables not targeted by the weighting process.

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Appendix A: AAPOR-Required Disclosure Elements

This study analyzes three existing data sources and collected no new data. For each source, the elements required by the AAPOR Code are given below. Where an element is described in permanently archived, study-specific documentation, we link to it; where an element is not documented or does not apply, we say so and why. Year-specific details for the CES are collected in Table [A1](#).

Data Source 1: Cooperative Election Study (CES), 2006–2022

Data source: Survey data. Secondary analysis of the publicly released CES Common Content for the nine federal election years 2006 through 2022. Data and documentation for each year are archived on Harvard Dataverse: 2006, <https://doi.org/10.7910/DVN/Q8HC9N>; 2008, <https://doi.org/10.7910/DVN/YUYIVB>; 2010, <https://doi.org/10.7910/DVN/VKKRWA>; 2012, <https://doi.org/10.7910/DVN/HQEVPK>; 2014, <https://doi.org/10.7910/DVN/XFXJVY>; 2016, <https://doi.org/10.7910/DVN/GDF6Z0>; 2018, <https://doi.org/10.7910/DVN/ZSBZ7K>; 2020, <https://doi.org/10.7910/DVN/E9N6PH>; 2022, <https://doi.org/10.7910/DVN/PR4L8P>.

Data collection strategy: Online surveys, with a pre-election and a post-election wave in each year. In 2006, additional profile information was collected in a separate survey in August 2006.

Research sponsor and conductor: The CES is sponsored by a consortium of universities and research teams, with support from the National Science Foundation. Data were collected by Polimetrix (2006), YouGov/Polimetrix (2008–2012), and YouGov (2014–2022). No funding was provided for this study.

Measurement tools/instruments: The exact wording, response options, and instructions for every question analyzed are in each year’s questionnaire and guide, archived with the data

above. Table [A2](#) lists, for each benchmarked item and year, the CES variable used, verified against the variable labels in the released data files. In 2008 the union item (CC329) asks about union membership in the household rather than the respondent’s own membership. Vote-choice items asked respondents to choose among parties in earlier years and among named candidates, shown with party labels, in later years; see “Measuring Error” in the main text.

Population under study: The CES describes its samples as representative of U.S. citizens ages 18 and older, the population from which its sampling frames are drawn. Estimates are evaluated for the 50 states and the District of Columbia and, for U.S. House, congressional districts, in each election year. Respondents were not excluded on citizenship; in the years that include a direct citizenship item (`cit1`), 1,134 (2018), 1,010 (2020), and 624 (2022) respondents reported not being U.S. citizens. CPS benchmarks, by contrast, are restricted to citizens (Data Source 2).

Methods used to generate and recruit the sample: The samples are non-probability samples drawn from opt-in online panels: the Polimetrix/YouGov panel (PollingPoint through 2012) and, from 2008, panels supplied by other companies (Tables [5](#) and [A1](#)). Coverage is limited to people who joined those panels.

In every year, a target sample was drawn from a sampling frame and matched to completed interviews from the panels by sample matching (Table [3](#)). In 2006, the target population was enumerated from commercial consumer databases, and the target sample was stratified by registration status, state size, and district competitiveness (16 strata), and further by age, race, and gender; more than 150,000 PollingPoint respondents formed the pool from which matches were drawn, and Polimetrix used multiple matching and rematching to handle nonresponse. From 2008 through 2020, frames were constructed by stratified sampling from American Community Survey (ACS) citizen records, with political and religious variables attached from the Current Population Survey (CPS), exit polls, and the Pew Religious Landscape Survey

as documented in each guide. In 2022, the frame was a “politically representative modeled frame” based on the 2019 ACS, with political variables modeled from voter files, the 2020 CPS Voting and Registration Supplement, the 2020 National Election Pool exit poll, and the 2020 CES.

Panelists were selected for invitation using a five-way cross-classification (age $\times$ gender $\times$ race $\times$ education $\times$ state) from 2008 through 2016, and a six-way cross-classification (age $\times$ gender $\times$ race $\times$ education $\times$ region $\times$ sample source) from 2018. No other quotas are documented. All pre-election respondents were re-invited to the post-election wave. The guides do not describe the incentives offered to panelists, except that in 2006 “inducements” were offered late in October to increase participation among low-income and low-interest respondents; YouGov currently describes its panel incentive as points redeemable for small rewards (YouGov 2026), and incentives offered by external providers are not documented. Data collection was not in person.

Method(s) and mode(s) of data collection: Self-administered web questionnaires. The guides do not report the languages in which the questionnaires were offered.

Dates of data collection: Field periods for each year are in Table 2. Guides through 2014 report them only at monthly or relative precision.

Sample sizes and discussion of precision: Released sample sizes, the number of completed pre-election interviews in the matching pool, and post-election completes are in Tables 2 and A1. Because the samples are not probability samples, we report no margins of sampling error; this study instead measures total error directly, as the RMSE of CES estimates against external benchmarks. The trend analysis (Table 13) reports model-based HC3 standard errors and permutation p -values; these describe uncertainty in the fitted trend across variable-years, not sampling error in the CES estimates.

Whether and how the data were weighted: Estimates are analyzed unweighted (matching only), with the CES-provided weights, and with anesrake calibration weights we con-

structed. The CES weights were computed by weighting the matched sample to the frame (propensity-score subclassification in 2008–2012; entropy balancing from 2014) and then post-stratifying (raking in 2022), using the variables in Table 3; weights were trimmed at 7 (2008–2012) or 15 (2014–2022) and normalized to the sample size (see “CES Sample Matching and Weighting Procedure” and Appendix C). The CES weight variable used in each year is listed in Table A1. Where the CES provides a separate post-election weight (2016–2022), that weight is applied to every item, including demographics measured in the pre-election wave, so that all estimates in a year rest on the same respondents and weights. In 2014 the released file contains a single weight. The anesrake weights, their targets (CPS and administrative benchmarks), and their parameters are described in “Weighting Scenarios” and Table 7.

How the data were processed and procedures to ensure data quality: The guides state that pre-election interviews entered the matching pool “after quality controls were applied” but do not describe those controls. Our processing, including recoding CES responses to benchmark categories, is described in “Measuring Error.” Responses of “don’t know,” “not sure,” “did not vote in this race,” and “I did not vote,” along with skipped, not-asked, and no-race responses, were set to missing and dropped item by item; “other” responses were retained in the denominator. No filter on reported turnout was applied to vote-choice items. State-level estimates from 2008 onward use the state recorded in the post-election wave, so respondents who completed only the pre-election wave are excluded from all state-level estimates; the 2006 estimates use the pre-election state variable (v1002). U.S. House estimates exclude respondents without a post-election congressional district; in 2020, a further 774 North Carolina respondents flagged in the CES data were excluded from House estimates. In 2018, respondents in eight states (Texas, Pennsylvania, Indiana, South Carolina, Alabama, Kentucky, Oklahoma, and Utah) who reported voting a straight party ticket (CC18_408 = 2; $n = 4,021$) were not asked about individual statewide and U.S. House contests. Following the procedure described in the 2018 guide, we assigned each of these respondents the candidate of their ticket’s party for U.S. Senate, governor, U.S. House, attorney general, and secretary

of state; respondents whose ticket party fielded no candidate, who reported a minor-party ticket, or whose ticket party was not recorded were left missing. Their state legislative vote choices were taken directly from CC18.413c and CC18.413d. Also in 2018, the post-election questionnaire displayed Kansas’s attorney general and secretary of state candidates to Kentucky respondents, where neither office was on the ballot, and displayed no candidates to Kansas respondents, none of whom has a recorded response to either item. The 2018 guide does not mention this error. Kentucky has no benchmark for these offices and Kansas has no responses, so both states are absent from the 2018 estimates for these offices.

CES candidate names were matched to official returns first by exact, case-insensitive comparison and otherwise by Jaro string distance (R package `stringdist` 0.9.15, method "jw" with $p = 0$), taking the closest candidate with no threshold. Of 3,612 candidate contests, 2,530 matched exactly and 1,082 by string distance (maximum distance 0.48). All 1,082 distance-based matches were reviewed manually; one was overridden, and party affiliations were assigned manually where official returns omitted them (Appendix D).

Panel description: Panel recruitment is described in “Sourcing Panel Members” in the main text. Procedures for managing panel membership, participation, and attrition are not described in the CES guides.

Interviewer details: No interviewers were used; the questionnaires were self-administered. Human coding in this study was limited to the manual review of distance-based candidate matches, one override, and party assignments described above, all performed by the authors.

Screening criteria and process: The frames were restricted to U.S. citizens ages 18 and older. The guides do not document any other eligibility screening or whether respondents were routed from other surveys.

Study stimuli: None beyond the questionnaires archived with the data.

Dispositions or response or participation rates: Case dispositions and the rates the

guides label AAPOR Response Rate 1 are reported in Table [A1](#) for 2008–2022; the 2006 guide reports none. The reporting base differs across years: 2014 and 2016 report YouGov and external-panel cases separately and combined, 2018 reports YouGov panelists only, and the remaining years do not specify. Because the number of people exposed to recruitment is unknown, these figures are completion rates among invited panelists, not response or participation rates for the population (see the main text); a participation rate for the population cannot be computed.

Sample sizes: The number of respondents underlying each estimate is shown in Figure [4](#), respondents per state in Table [1](#), and the number of estimates by variable and year in Table [6](#). Variable-years excluded from particular analyses, and why, are stated in the main text and table notes.

Measurement and model specification: RMSE is defined in “Measuring Error.” The trend models are specified in Table [13](#) and Figure [3](#), the sample-size model in Figure [5](#), and the anesrake weights in “Weighting Scenarios.”

Limitations: The CES samples are non-probability samples, so their error cannot be characterized by sampling theory. Turnout, voting method, and vote choice are self-reports, so measured error includes misreporting as well as sample composition. The turnout benchmark uses the voting-age population as its denominator, which includes noncitizens and others ineligible to vote, whereas the CES population is adult citizens. CES estimates include self-reported noncitizens, whereas CPS benchmarks are restricted to citizens. Documentation is incomplete for some waves: the 2006 guide does not describe a weighting procedure, the 2012 guide repeats the 2010 description, several guides repeat identical figures for the size of the matching pool, and no guide describes the quality controls applied. Some benchmark outcomes (U.S. Senate and gubernatorial results from 2014 onward, and presidential vote in 2020) were used in the CES weights, so weighted estimates for them are not independent tests of accuracy.

Table A1: CES Disclosure Details by Year

Year	Panels	Dispositions of invited cases (I / P / R / UH / NE)	RR1	Post-election completes	CES weight used
2006	PollingPoint	Not reported (>150,000 in matching pool)	–	28,757	v1001
2008	PollingPoint; E-Rewards; Western Wats	49,043 / 13,137 / 627 / 42,007 / 1,082	.468	27,021	V201
2010	PollingPoint; E-Rewards; Western Wats	75,450 / 27,155 / 4,645 / 79,723 / 9,262	.404	46,684	V101
2012	PollingPoint; E-Rewards; Western Wats	83,583 / 20,229 / 7,754 / 138,624 / 14,267 ^a	.334 ^a	45,018 ^d	V103
2014	YouGov; MyPoints; Research Now; SSI	YouGov: 43,203 / 6,480 / 1,897 / 29,448 / 1,442 External: 52,383 / 28,773 / 7,392 / 238,639 / 19,643	.533 / .160 / .234 ^b	48,853	weight
2016	YouGov; MyPoints; Research Now; SSI; GMI	YouGov: 53,939 / 7,177 / 2,519 / 64,997 / 982 External: 52,443 / 22,692 / 5,210 / 557,367 / 43,822	.419 / .082 / .139 ^b	52,899	commonweight_post
2018	YouGov; Dynata; Critical Mix; Prodege	YouGov only: 63,301 / 8,341 / 125 / 139,374 / –	.300 ^c	51,808	commonpostweight
2020	YouGov; Dynata; Critical Mix; Prodege	60,466 / 7,900 / 1,591 / 28,922 / 1,606 (74,099 in matching pool)	.612	51,551	commonpostweight
2022	YouGov; Dynata; Disqo; Prodege; Generation Lab	64,957 / 10,771 / 3,102 / 57,676 / 3,017 (80,233 in matching pool)	.476	50,981	commonpostweight

Note: I = complete; P = partial; R = refusal and breakoff; UH = unknown eligibility; NE = not eligible. Dispositions and RR1 are as reported in each year's CES guide; RR1 is reported in the main text as a completion rate. ^aCommon Content sample only; the 2012 guide also reports separate panel and NES samples. ^bYouGov / external / combined. ^cYouGov panelists only. The 2014, 2016, and 2018 guides each give the matching pool as "approximately 87,389"; because the figure is identical across years and the 2016 disposition table reports 106,382 completes, we do not report it. Post-election completes are counts of respondents coded as completing the post-election wave in each year's data file (v4001, V400, V403, or tookpost). ^dThe 2012 file has no post-election disposition variable; the count is the number of respondents with a non-missing post-election state (inputstate_post), the variable used to assign respondents to states.

Table A2: CES Variables Used for Each Benchmarked Item, by Year

Item	2006	2008	2010	2012	2014	2016	2018	2020	2022
Age Group	v2020 ^a	V207 ^a	V207 ^a	birthyr ^a	birthyr ^a	birthyr ^a	birthyr ^a	birthyr ^a	birthyr ^a
Sex	v2004	V208	V208	gender	gender	gender	gender	gender	gender4
Education	v2018	V213	V213	educ	educ	educ	educ	educ	educ
Hispanic Origin	—	—	V290	hispanic	hispanic	hispanic	hispanic	hispanic	hispanic
Employment Status	v2030	V209	V209	employ	employ	employ	employ	employ	employ
Family Income	v2032	V246	V246	faminc	faminc	faminc	faminc_new	faminc_new	faminc_new
Union Membership	v2082	CC329	V264	union	union	union	union	union	union
Veteran Status	v3083	CC328.3	V254	milstat.3	milstat.3	milstat.3	milstat.3	milstat.3	milstat.3
Residence Duration	—	CC334	CC351	CC351	CC351	CC16.361	—	CC20.361	CC22.361
Voting Turnout	v4004	CC403	CC401	CC401	CC401	CC16.401	CC18.401	CC20.401	CC22.401
Voting Method	v4006	CC405	CC403	CC403	CC403	CC16.403	CC18.403	CC20.403	CC22.403
President	—	CC410	—	CC410a	—	CC16.410a	—	CC20.410	—
U.S. Senate	v4014	CC411	CC410a	CC410b	CC410b	CC16.410b	CC18.410b	CC20.411	CC22.411
U.S. House	v4015	CC412	CC412	CC412	CC412	CC16.412	CC18.412	CC20.412	CC22.412
Governor	v4013	CC413	CC411	CC411	CC411	CC16.411	CC18.411	CC20.413	CC22.413
Attorney General	v4017	—	CC413a	CC413a	CC413a	CC16.413a	CC18.420a	CC20.414a	CC22.414a
Secretary of State	v4018	—	CC413b	CC413b	CC413b	CC16.413b	CC18.420b	CC20.414b	CC22.414b
State Senator	v4021	CC414.2	CC413c	CC413c	CC413c	CC16.413c	CC18.413c	CC20.415c	CC22.415c
State Representative	v4020	CC414.1	CC413d	CC413d	CC413d	CC16.413d	CC18.413d	CC20.415d	CC22.415d
State assignment	v1002 ^b	V259	V206_post	inputstate_post	inputstate_post	inputstate_post	inputstate_post	inputstate_post	inputstate_post
Congressional district	v1003	V264	V276_post	cdid113_post	cdid113_post	cdid115_post	cdid116_post	cdid116_post	cdid118_post

Note: Variable names are as they appear in each year’s released Common Content file and were verified against the variable labels in those files; question wording is in the corresponding guide and questionnaire on Dataverse. Dashes indicate that the item was not measured, or the office was not on the ballot, that year. Vote-choice items are from the post-election wave. ^aAge group is derived from year of birth. ^bThe 2006 file has no post-election state variable; the pre-election state is used. In 2008, CC329 asks about union membership in the household, not the respondent’s own membership.

Data Source 2: Current Population Survey (CPS)

Data source: Survey data. Secondary analysis of the CPS, obtained through IPUMS CPS (Flood et al. 2024), used for demographic and voting-method benchmarks.

Data collection strategy, sponsor, and conductor: A monthly household survey sponsored by the U.S. Census Bureau and the Bureau of Labor Statistics and conducted by the Census Bureau. Methodology, questionnaires, and sampling are documented at <https://www.bls.gov/cps/documentation.htm> and <https://cps.ipums.org/cps/>.

Measurement tools/instruments: Benchmarks come from the CPS basic monthly files, with all twelve months pooled within each year (the March file used is the basic monthly March sample, not the Annual Social and Economic Supplement), obtained through the IPUMS API (IPUMS CPS version 13.0). Sex, education, age, Hispanic origin, veteran status, employment status, and family income are from the basic monthly items; union membership is from the earnings (outgoing rotation) items; residence duration and voting method are from the November Voting and Registration Supplement. Questionnaires are documented at

the links above.

Population, sample, mode, and dates: A monthly probability sample of U.S. households, interviewed in person or by telephone, as described in “Benchmarks” in the main text; data from 2006 through 2022 were used. Benchmarks are restricted to respondents ages 18 and older who are U.S. citizens (IPUMS CITIZEN codes 1–4).

Sample sizes, response rates, and weighting: Annual sample sizes and household response rates are in Table B2; sample sizes count person-month records, not unique persons. Benchmarks use the CPS weights, which account for sampling design, nonresponse, and undercoverage (WTFINL for basic monthly items, EARNWT for union membership, and VOSUPPWT for Voting and Registration Supplement items).

Other elements: Interviewer procedures, screening, processing, and panel rotation are documented by the Census Bureau and BLS at the links above.

Limitations: CPS benchmarks carry their own sampling and nonsampling error, including proxy reports and, for voting method, self-report error; they are treated here as fixed, so part of the measured CES error may reflect benchmark error, particularly in small states. Race was not evaluated because CPS race categories could not be mapped exhaustively onto CES categories; Hispanic origin, measured comparably in both, was evaluated from 2010 onward.

Data Source 3: Election Returns and Turnout

Data source: Government records. Official election returns, compiled by the MIT Election Data and Science Lab, the Federal Election Commission, CQ Press, Ballotpedia, and state election offices, and turnout compiled by the University of Florida Election Lab (Table B4).

Measurement: Turnout is total ballots counted divided by the voting-age population (VAP), from the United States Elections Project (<https://www.electproject.org>). In 22 state-years for which total ballots are not reported, the numerator is votes cast for the highest

office on the ballot: Alabama 2006 and 2012; Connecticut 2008; Delaware 2012; Maine 2006; Mississippi 2006–2014; Missouri 2012; New Mexico 2014; Oklahoma 2006 and 2012; Pennsylvania 2006 and 2012; Rhode Island 2012; Texas 2006, 2008, 2012, and 2014; and West Virginia 2012. National turnout is summed ballots divided by summed VAP across the 50 states and the District of Columbia. State-level turnout benchmarks cover the 50 states; the District of Columbia is excluded. Vote shares are each candidate’s share of all votes cast in the contest. Each contest is compared on the candidate who received the most votes, except in eight contests whose winner was not among the candidates offered in the CES item (independent or write-in winners, and the 2016 Louisiana Senate race); there the major-party nominee offered by the CES is used instead. These contests are listed in Appendix D.

Processing: Matching of candidates and parties to CES responses, and the handling of multi-party endorsements and special elections, are described in Appendix D.

Elements that do not apply: These are official counts rather than samples, so sampling, mode, response rates, weighting, interviewers, and sampling precision do not apply.

Appendix B: Supplementary Tables and Figures

Table B1: National ANESRake-Weighted RMSE for Variables Used in Weighting

Variable	2006	2008	2010	2012	2014	2016	2018	2020	2022	Mean
ANESRake Full Variables										
<i>Demographic</i>										
Age Group	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Sex	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Hispanic Origin	-	-	0.1	0.0	0.0	0.0	0.0	0.1	0.0	0.0
Education	0.0	0.0	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.1
Family Income	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Employment Status	0.0	0.0	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.1
Union Membership	0.0	-	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Veteran Status	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Residence Duration	-	0.0	0.1	0.0	0.0	-	-	0.0	0.0	0.0
Demographic Mean	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0
<i>Voting Administration</i>										
Turnout	0.0	0.0	0.4	0.0	0.0	0.0	0.0	0.5	0.0	0.1
Voting Method	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Voting Administration Mean	0.0	0.0	0.3	0.0	0.0	0.0	0.0	0.2	0.0	0.1
Yearly Mean (Full)	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.1	0.0	0.0
ANESRake Restricted Variables										
<i>Demographic</i>										
Age Group	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Education	0.0	0.0	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.1
Family Income	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Employment Status	0.0	0.0	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.1
Demographic Mean	0.0	0.0	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Yearly Mean (Restricted)	0.0	0.0	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0

- Variable not used as an ANESRake target that year: either not provided or provided but not selected (see Table 7).

Note: RMSE in percentage points. National-level errors. ANESRake Full uses all available demographic and voting administration variables. ANESRake Restricted uses a subset (Age Group, Education, Family Income, Employment Status).

Table B2: CPS Sample Description, 2006–2022

Year	Sample Size	Probability Sample?	Response Rate	Field Dates
2006	1,221,832	Yes	91.7%	Monthly
2008	1,209,862	Yes	91.7%	Monthly
2010	1,230,780	Yes	92.0%	Monthly
2012	1,215,614	Yes	90.4%	Monthly
2014	1,218,742	Yes	88.9%	Monthly
2016	1,202,067	Yes	86.7%	Monthly
2018	1,150,355	Yes	84.4%	Monthly
2020	1,001,837	Yes	74.3%	Monthly
2022	964,927	Yes	72.9%	Monthly

Source: U.S. Bureau of Labor Statistics, series LNU09300000. <https://data.bls.gov/timeseries/LNU09300000>

Note: Sample sizes are yearly totals across all monthly surveys. Response rates are annual averages of monthly CPS household response rates.

Table B3: U.S. House District-Level RMSE by Sample Size and Race Competitiveness

Min. Respondents	Winner's Vote Share (%)				
	20–40	40–60	60–80	80–90	90–100
<i>RMSE (%)</i>					
All	10.0	8.8	9.4	11.0	10.3
30+	10.0	8.7	9.2	11.1	9.8
60+	4.3	8.3	8.6	10.3	10.5
90+	-	7.6	7.4	10.6	12.5
120+	-	7.1	6.9	9.6	16.6
<i>Number of District-Years</i>					
All	5	1037	1491	157	148
30+	5	1032	1465	151	141
60+	3	901	1237	106	86
90+	0	565	615	38	27
120+	0	190	181	7	3

Note: Each cell reports RMSE (in percentage points) for all congressional district-years (2008–2022) meeting both the respondent threshold (row) and winner's vote share range (column). A district-year is one district in one election year, so the same district recurs across waves. RMSE is pooled over all qualifying observations rather than averaged across waves, so larger waves count for more.

Table B4: Data Sources

Category	Source	URL
Presidential	MIT Election Data and Science Lab	https://electionlab.mit.edu/research
U.S. Senate & House	Federal Election Commission	https://www.fec.gov/introduction-campaign-finance/election-results-and-voting-information/
Gubernatorial	CQ Press Elections Collection	https://library.cqpress.com/elections/
State legislative/executive	Ballotpedia	https://ballotpedia.org/Election_results
Demographics	IPUMS CPS (U.S. Census Bureau)	https://cps.ipums.org/cps/
Turnout	University of Florida Election Lab	https://election.lab.ufl.edu/research/voter-turnout/
Voting method	IPUMS CPS	https://cps.ipums.org/cps/

Note: Data cover federal election years 2006–2022.

Table B5: RMSE Before and After CES Weighting by Variable Class and Weighting Status

Panel and Estimate	2008	2010	2012	2014	2016	2018	2020	2022	Mean
A. Primary Demographic Variables									
Matching-Only	5.2	9.8	7.1	5.5	7.4	6.4	7.1	7.0	7.0
Matching + Weighting	4.9	5.6	3.0	3.5	3.3	3.5	3.4	3.3	3.8
Δ RMSE	-0.4	-4.3	-4.1	-2.0	-4.1	-3.0	-3.7	-3.7	-3.1
B. Secondary Demographic Variables									
Matching-Only	5.5	8.1	8.5	7.0	6.3	6.7	6.8	7.1	7.0
Matching + Weighting	5.7	6.4	7.0	6.5	6.6	6.8	7.5	7.4	6.7
Δ RMSE	+0.1	-1.7	-1.4	-0.5	+0.3	+0.1	+0.7	+0.3	-0.3
C. Primary Candidate Choice Variables									
Matching-Only	–	–	–	5.3	7.2	5.8	10.4	10.4	7.8
Matching + Weighting	–	–	–	3.3	4.0	4.3	3.8	5.3	4.1
Δ RMSE	–	–	–	-2.0	-3.2	-1.5	-6.6	-5.1	-3.7
D. Secondary Candidate Choice Variables									
Matching-Only	6.4	6.0	3.6	–	–	–	–	–	5.3
Matching + Weighting	4.9	2.7	2.9	–	–	–	–	–	3.5
Δ RMSE	-1.5	-3.3	-0.7	–	–	–	–	–	-1.8

Note: RMSE is reported in percentage points. Δ RMSE is calculated as matching + weighted RMSE minus matching-only RMSE. Negative values indicate that weighting reduced error; positive values indicate that weighting increased error. Each variable receives equal weight when calculating a panel's yearly mean.

Panel A includes Age Group, Education, and Sex. Panel B includes Employment Status, Family Income, Union Membership, and Veteran Status.

Panels C and D include Governor and U.S. Senate. These variables were secondary from 2008 through 2012 and primary from 2014 through 2022. Dashes indicate that the variables did not have the panel's classification in that year, rather than missing data.

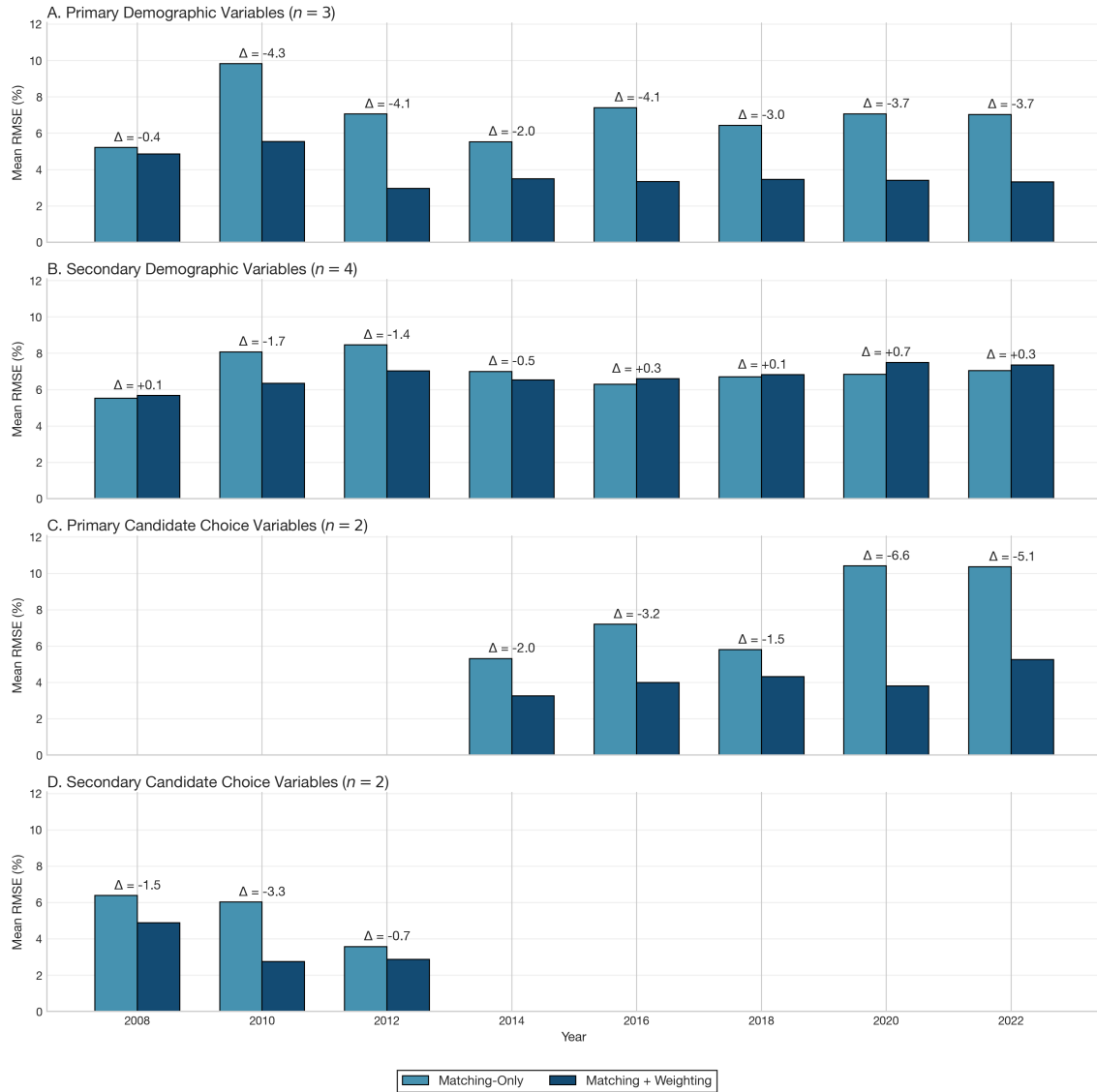


Figure B1: Weighting and RMSE by Variable Type

Bars compare matching-only and weighted RMSE. Δ is weighted minus matching-only RMSE, so negative values indicate improvement. Panels A and B contain primary demographics (Age Group, Education, and Sex) and secondary demographics (Employment Status, Family Income, Union Membership, and Veteran Status), respectively. Panels C and D contain Governor and U.S. Senate, which were secondary in 2008–2012 and primary in 2014–2022; blank years indicate the other classification. Weighting improves primary demographics more than secondary demographics and produces larger reductions for the candidate-choice variables during their primary years, although the latter comparison is confounded with time.

Alt text: Four panels of paired bars comparing error before and after weighting by survey year, from 2008 to 2022. Weighting lowered error for primary demographic variables in every year, and for Governor and United States Senate in every year shown, both when they were weighting targets and when they were not. For secondary demographic variables, error changed little, and weighting slightly raised it in five of eight years.

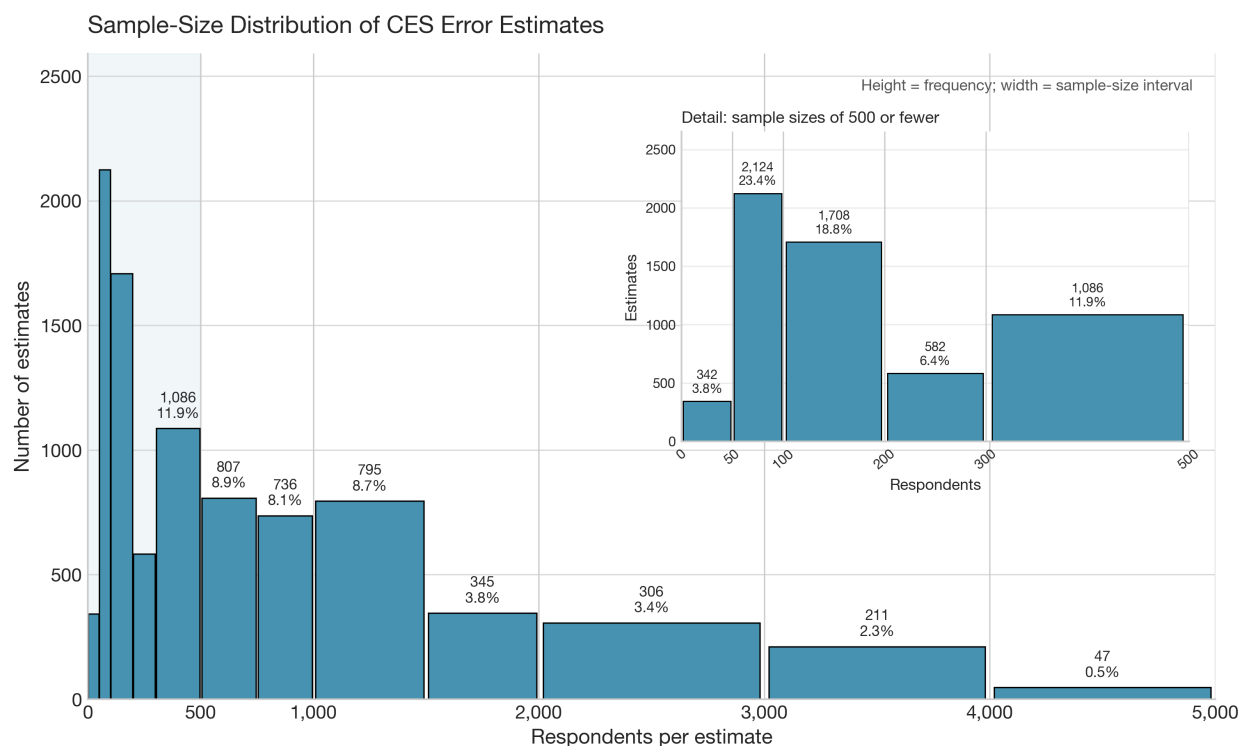


Figure B2: Distribution of Sample Sizes Underlying CES Error Estimates

Sample size is the number of CES respondents contributing to the survey estimate whose error is measured against an external benchmark. For statewide estimates, this is the relevant state-year respondent count; for U.S. House, which is a district-specific estimate, it is the relevant district-year respondent count. Bars represent unequal sample-size intervals: their heights report the number of estimate-level observations in each interval, their widths reflect the range of sample sizes covered, and their labels report the percentage of all included estimates. The figure includes unweighted CES estimates and excludes turnout (450 estimates), so the bars total 9,090 rather than the 9,540 in Figure 4.

Alt text: Histogram with bars of unequal width showing how many error estimates, excluding turnout, fall in each sample size range up to 5,000 respondents, with an enlarged inset for ranges up to 500. Nearly two-thirds of estimates rest on fewer than 500 respondents, and the 51 to 100 range is the largest, with 23.2 percent.

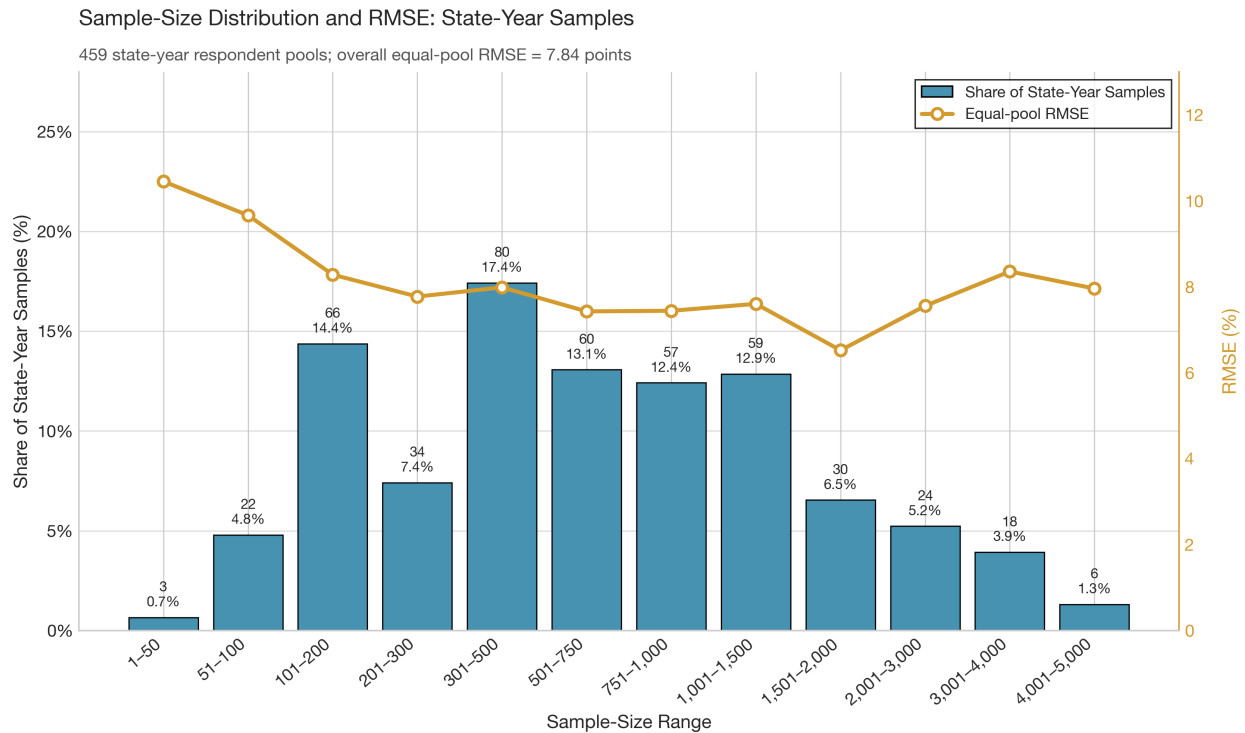


Figure B3: Sample-Size Distribution and RMSE Across State-Year Respondent Pools

This figure provides a sample-level counterpart to the estimate-level sample-size distribution reported in Figure 4. In that figure, each variable-specific comparison with an external benchmark is an error observation. Consequently, the same underlying state-year respondent pool appears repeatedly because its respondents are used to estimate multiple variables and, where applicable, multiple district-specific outcomes. Here, each state-year respondent pool appears only once. Its sample size is defined as the largest valid respondent count observed for that state-year; smaller variable-specific counts may result from item nonresponse or geographically restricted estimates. Bar heights report the percentage of state-year pools in each sample-size range, and labels report the number and percentage of pools. RMSE is calculated hierarchically so that variables producing more observations and pools producing more error rows do not receive disproportionate weight. Squared percentage-point errors are first averaged within each state-year-variable combination. Those variable-level mean squared errors are then averaged within each state-year pool, giving every variable equal weight. Within each sample-size range, the pool-level mean squared errors are averaged across state-year pools and the square root is taken, giving every pool equal weight. The overall equal-pool RMSE shown in the figure is calculated analogously across all state-year pools. Results use unweighted CES estimates and exclude turnout to match the corresponding main-text analysis.

Alt text: Bar chart with an overlaid line for 459 state and year respondent pools. Bars show the share of pools in each sample size range, peaking at 301 to 500 respondents. The line shows error for each range: about 10 percentage points for the smallest pools, and between 6.4 and 8.3 points for pools above 100 respondents, with no steady decline as pools grow.

Appendix C: Entropy Balancing Procedure

Beginning in 2014, the Cooperative Election Study (CES) weighted the matched sample to the target sample using entropy balancing (Hainmueller 2012). Entropy balancing is a multivariate reweighting method that assigns scalar weights to individual observations such that the reweighted sample exactly satisfies a set of covariate balance constraints while remaining as close as possible to a set of base weights.

Formally, weights were obtained by minimizing Kullback–Leibler divergence (Kullback 1959) between the final weights w_i and base weights q_i :

$$\min H(w) = \sum_i w_i \log\left(\frac{w_i}{q_i}\right),$$

subject to the following constraints:

$$\sum_{i \in S} w_i c_{ri}(X_i) = 0, \tag{C1}$$

$$\sum_{i \in S} w_i = 1, \tag{C2}$$

$$w_i \geq 0 \quad \forall i. \tag{C3}$$

Here, S indexes the matched sample, w_i denotes the entropy-balanced weight assigned to respondent i , and q_i denotes the base weight. In settings where base weights are unavailable—as is typical in nonprobability samples—base weights were set uniformly such that $q_i = 1/N$, where N is the matched sample size.

Constraint (C1) enforces balance on the r -th moment of covariate X : each moment function is centered on the corresponding target-sample moment, so the weighted sum equals zero exactly when the weighted sample reproduces that moment. In the CES implementation,

moment functions were specified as

$$c_{ri}(X_{ij}) = X_{ij} - \mu_j,$$

where X_{ij} denotes the value of covariate j for respondent i , and μ_j is the target-sample mean of covariate j . Constraints (C2) and (C3) ensure that the resulting weights form a valid probability distribution.

Minimizing Kullback–Leibler divergence subject to these constraints yields a set of weights that deviate minimally from the base weights while achieving exact balance on the specified covariates. This property ensures that weights vary smoothly across observations and preserves their interpretation as probability weights.

Appendix D: Inconsistencies and Intricacies of Election Information Reporting

Candidate names occasionally differed in format between the CES questionnaire and official election returns. Jaro string distance (Jaro 1989) was used to map CES candidates to those in the election returns; where no exact match existed, the closest name was taken, and all matches were manually reviewed.

Some states did not report candidate party affiliation in official election returns, even when candidates were formally affiliated with a party. In these cases, candidates were manually assigned to their affiliated party to avoid artificially inflating CES error, as such candidates would otherwise have been treated as unaffiliated despite being identified by CES respondents as Democrats or Republicans.

In states such as New York and Vermont, candidates endorsed by multiple parties appeared multiple times on the ballot—once for each party line. For example, a candidate may have received votes under the Democratic Party, Working Families Party, and Progressive Party.

In these cases, all votes in the election returns were aggregated to the candidate’s major-party affiliation (Democratic or Republican).

When a state held a special election on the same day as a general election, those votes were included. When both a special and a general election for the same office occurred on the same day, only the general election was counted.

In eight contests, CES vote choice is compared with the benchmark share of a major-party nominee rather than that of the top vote-getter: U.S. Senate in Alaska (2010), Maine (2012 and 2018), Vermont (2012 and 2018), and Louisiana (2016); governor of Rhode Island (2010); and secretary of state of North Dakota (2018). In all but Louisiana, the winner was an independent or write-in candidate not offered in the CES item.

Appendix E: U.S. House Sample Size and Competitiveness

Table [B3](#) stratifies U.S. House races by the winner’s vote share and by minimum district-level respondent thresholds. Two patterns stand out. Among competitive and moderately uncompetitive races, RMSE generally declines as the minimum respondent threshold rises, consistent with district-level sample size contributing to House vote-choice error. In very safe districts, by contrast, error remains elevated and the respondent-count gradient weakens, suggesting that some of the error in lopsided races reflects factors other than finite-sample variability. The highest-threshold cells in the 90–100 percent column rest on very few district-years and should be read with caution.

Appendix F: Race Salience

We report a supplementary analysis examining whether CES accuracy varies with race salience. Prior work in election polling suggests that higher-salience contests are typically measured more accurately than down-ballot races. Although the CES captures retrospective vote reports rather than pre-election intentions, it is useful to assess whether a similar pattern

appears in these data.

To minimize potential confounds, the analysis used unweighted data measured at the party level. This specification avoids the influence of CES weights and ensures comparability across offices by holding the measurement scale constant.

Under this specification, RMSE was lowest for high-salience contests (7.4 percent). Error was higher for down-ballot offices, though the pattern was not strictly monotonic: medium-salience races had the highest average RMSE (10.4 percent), while low-salience offices averaged 9.8 percent. Figure F1 presents these results by salience category.

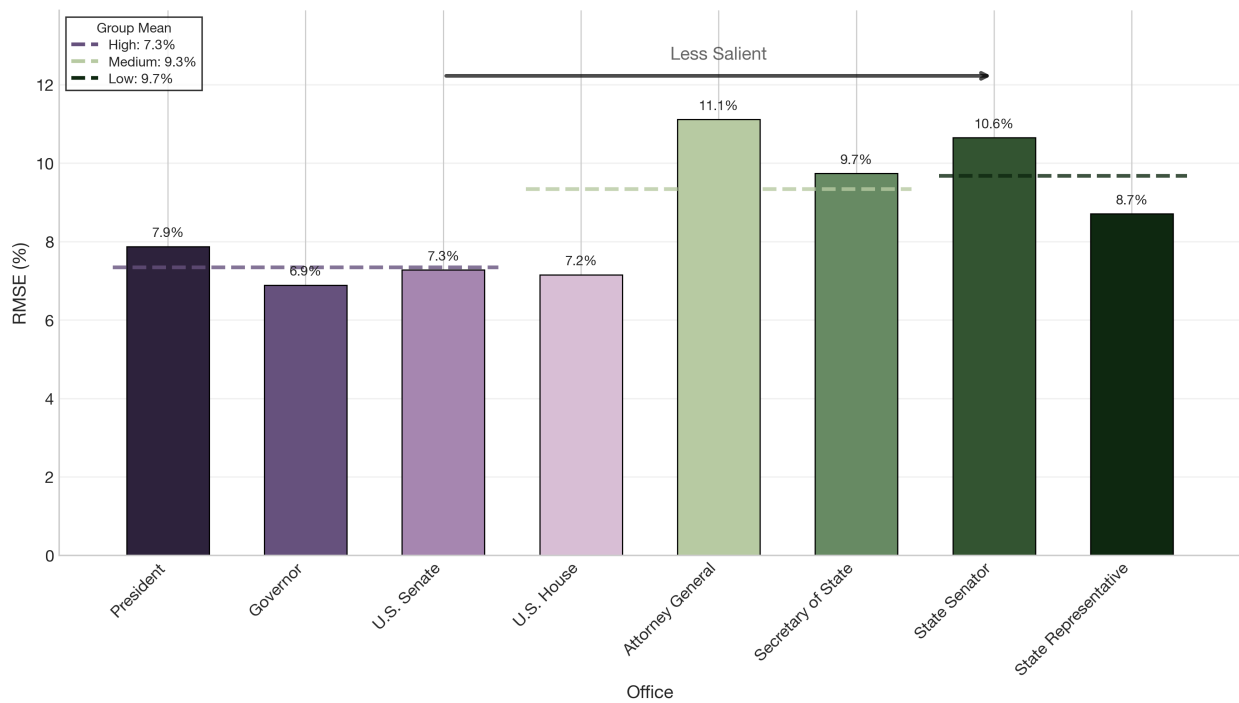


Figure F1: RMSE by Office Salience

RMSE is computed across all party-level, unweighted observations for each office, pooled over the 2006–2022 waves rather than averaged across them, so a wave contributes in proportion to its number of observations.

Alt text: Bar chart of error, in percentage points, for eight offices ordered from most to least salient, with dashed lines marking group averages. High salience offices (President, Governor, and United States Senate) average 7.4 points; medium salience offices (United States House, Attorney General, and Secretary of State) average 10.4; and low salience offices (State Senator and State Representative) average 9.8.